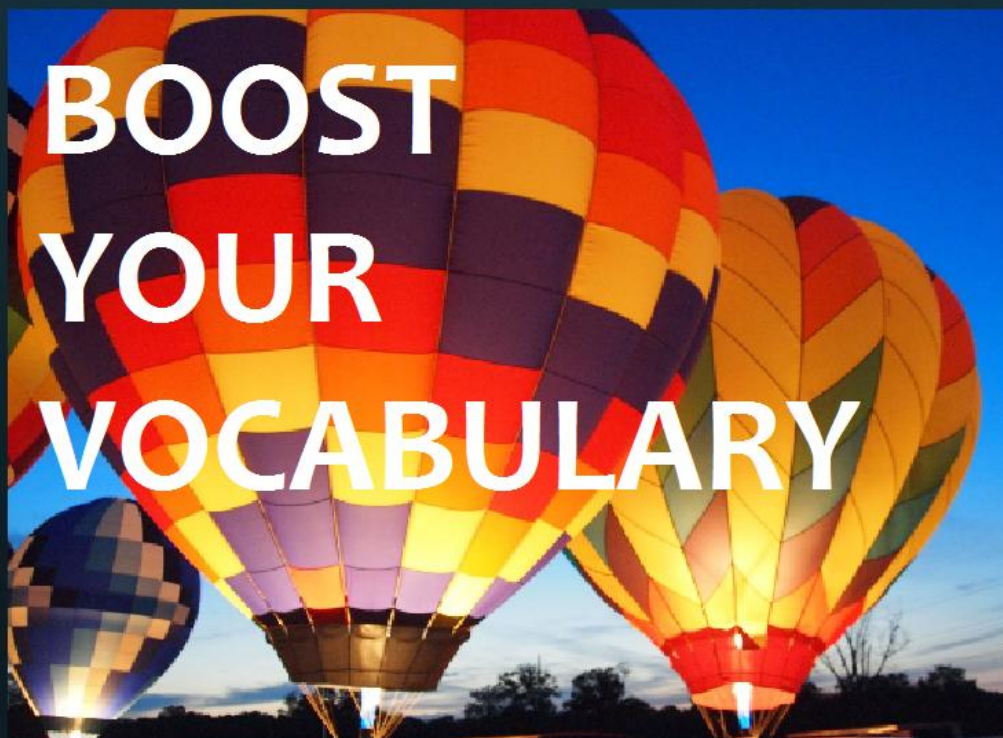


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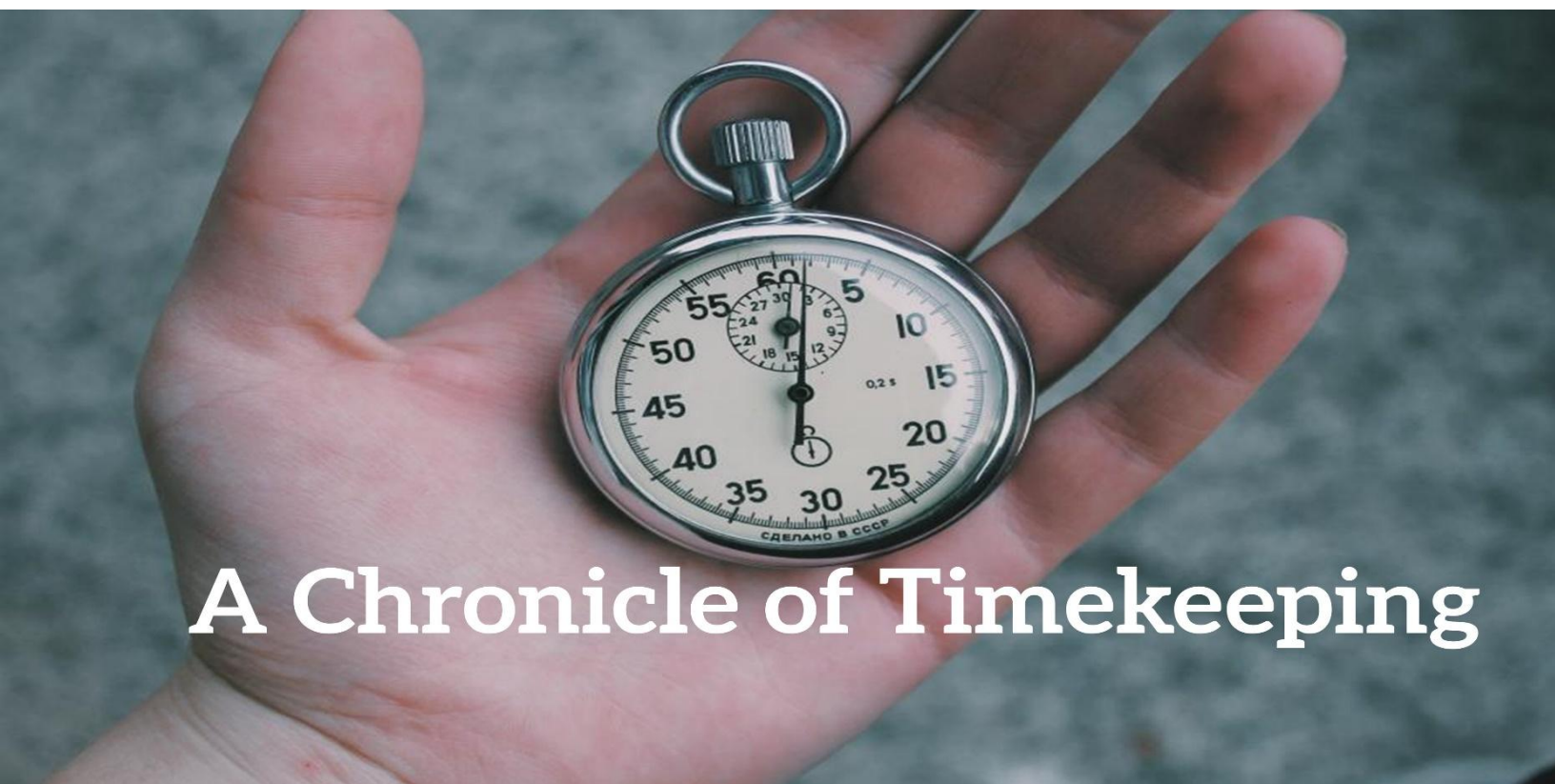
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8

TEST 1

READING PASSAGE 1



A Chronicle of Timekeeping

Our conception of time depends on the way we measure it.

A According to archaeological evidence, at least 5,000 years ago, and long before the **advent** of the Roman Empire, the Babylonians began to measure time, introducing calendars to **co-ordinate** **communal** activities, to plan the shipment of goods and, in particular, to regulate planting and harvesting. They based their calendars on three natural cycles: the **solar** day, marked by the successive periods of light and darkness as the earth rotates on its **axis**; the **lunar** month, following the phases of the moon as it **orbits** the earth; and

chronicle = a written record, history, story of historical events. 'krɒnɪkl̩
timekeeping = the activity of recording the time something takes 'taɪm ,ki:piŋ

advent = coming, start, arrival, the time when something first begins to be widely used. 'ædvənt
co-ordinate = organize, manage, direct, to make various, separate things work together. kəʊ'ɔ:dɪneɪt
communal = shared, common, public, relating or belonging to all the people living in a particular. 'kɒmjʊnəl
regulate = control, adjust, standardize. 'regjuleɪt
solar = relating to the Sun 'səʊlə
axis = alignment, centre line, (the imaginary line around which a large round object, such as the Earth). 'æksɪs
lunar = relating to the Moon. 'lu:nə
orbit = circle, revolve around, travel around, go around, 'ɔ:bɪt

the solar year, defined by the changing seasons that **accompany** our planet's revolution around the sun.

B

Before the invention of **artificial** light, the moon had greater social impact. And, for those living near the **equator** in particular, its **waxing and waning** was more **conspicuous** than the passing of the seasons. Hence, the calendars that were developed at the lower **latitudes** were influenced more by the lunar cycle than by the solar year. In more northern **climes**, however, where seasonal agriculture was practised, the solar year became more **crucial**. As the Roman Empire expanded northward, it organised its activity chart for the most part around the solar year.

C

Centuries before the Roman Empire, the Egyptians had **formulated** a **municipal** calendar having 12 months of 30 days, with five days added to approximate the solar year. Each period of ten days was marked by the appearance of special groups of stars called **decans**. At the rise of the star Sirius just before sunrise, which occurred around the all-important annual flooding of the Nile, 12 decans could be seen spanning the heavens. The **cosmic** significance the Egyptians placed in the 12 decans led them to develop a system in which each interval of darkness (and later, each interval of daylight) was divided into a dozen equal parts. These periods became known as **temporal hours** because their **duration** varied according to the changing length of days and nights with the passing of the seasons. Summer hours were long, winter ones short; only at the spring and autumn **equinoxes** were the hours of daylight and darkness equal. Temporal hours, which were first **adopted** by the Greeks and then the Romans, who **disseminated** them through Europe, remained in use for more than 2,500 years.

D

In order to **track** temporal hours during the day, inventors created **sundials**, which indicate time by the length or direction of the sun's shadow. The

accompany = go together with, come with, be associated with, happen with, appear with.
ə'kʌmpəni

artificial = man-made, synthetic, non-natural.
ɑ:'tʃɪʃl

equator = an imaginary line drawn around the middle of the Earth. i'kwetə

wax and wane = to increase and decrease over time. 'wæks ənd weɪn

conspicuous = obvious, clear, noticeable.
kən'spɪkjʊəs

latitude = the distance north or south of the equator, measured in degrees. 'lætɪtju:d

clime = zone, region, a place that has a particular type of climate. kləɪm

crucial = vital, fundamental, essential, important, necessary, key. 'kru:ʃl

formulate = invent, create, make, develop.
'fɔ:mjuleɪt

municipal = civic, public, community, #private.
mju:'nɪsɪpl

decans = The decans (Egyptian) are 36 groups of stars (small constellations) used in the Ancient Egyptian astronomy. 'dek(ə)n

cosmic = relating to space or the universe.
'kɒzmɪk

interval = intermission, interlude, break. 'ɪntəvəl

temporal hours = a unit of time used in the past that divided the daylight into an equal number of hours, 'tempərəl 'aʊəz

duration = the length of time that something lasts. dju'reɪʃn

equinox = solstice, one of the two times in a year when night and day are of equal length.
'i:kwɪnɒks

adopt = accept, approve, implement, apply, #reject ə'dɒpt

disseminate = spread, publish, distribute.
dɪ'semɪneɪt

track = follow, trace, pursue. træk

sundial = an object used in the past for telling the time. 'sʌndaɪəl

sundial's **counterpart**, the water clock, was designed to measure temporal hours at night. One of the first water clocks was a basin with a small hole near the bottom through which the water **dripped out**. The falling water level **denoted** the passing hour as it **dipped** below hour lines **inscribed** on the inner surface. Although these devices performed **satisfactorily** around the Mediterranean, they could not always be depended on in the cloudy and often freezing weather of northern Europe.

E

The advent of the mechanical clock meant that although it could be adjusted to maintain temporal hours, it was naturally suited to keeping equal ones. With these, however, **arose** the question of when to begin counting, and so, in the early 14th century, a number of systems **evolved**. The **schemes** that **divided** the day into 24 equal parts varied according to the start of the count: Italian hours began at sunset, Babylonian hours at sunrise, **astronomical** hours at midday and 'great clock' hours, used for some large public clocks in Germany, at midnight. Eventually these were **superseded** by 'small clock', or French, hours, which split the day into two 12-hour periods **commencing** at midnight.

F

The earliest recorded **weight-driven mechanical clock** was built in 1283 in Bedfordshire in England. The revolutionary aspect of this new timekeeper was neither the **descending** weight that provided its motive force nor the gear wheels (which had been around for at least 1,300 years) that transferred the power; It was the part called the **escapement**. In the early 1400s came the invention of the coiled spring or fusee which maintained constant force to the gear wheels of the timekeeper despite the changing tension of its **mainspring**. By the 16th century, a **pendulum** clock had been devised, but the pendulum swung in a large arc and thus was not very efficient.

counterpart = equal, colleague, equivalent.

'kauntəpɑ:t

drip = drop, come out, leak, #stream. drɪp

denote = indicate, represent, refer to, #connote
dɪ'neɪt

dip = dunk, immerse, to put something into a liquid for a very short time and take it out again.

dɪp

inscribed = engrave, carve, to carefully cut, print or write on smt ɪn'skraɪbd

satisfactory = pleasing, reasonable, acceptable, adequate, #unsatisfactory sətɪs'fæktəri

arise = rise, ascend, appear, # retire ə'raɪz

evolve = change, grow, advance, to develop and change gradually over a long period of time. ɪ'vɒlv

scheme = plan, idea, method. ski:m

divide = split, separate, distribute, allocate, #join. dɪ'vaɪd

astronomical = relating to the scientific study of the stars. æstrə'nɒmɪkəl

supersede = replace, supplant, displace.
su:pə'si:d

commence = start, begin, originate. kə'mens

weight-driven mechanical clock = a clock using a pendulum weɪt - 'drɪvən mɪ'kænɪkəl 'klɒk

descend = downward, fall, drop, go down. dɪ'send

escapement = a piece of machinery in a clock from the spring or weight to a wheel. ɪ'skeɪpmənt

mainspring = the most important spring in a watch or clock. 'meɪnsprɪŋ

pendulum = a long metal stick with weight at the bottom that swings regularly from side to side to control the working of a clock. 'pendjʊləm

G

To address this, a variation on the original escapement was invented in 1670, in England. It was called **the anchor escapement**, which was a lever-based device shaped like a ship's anchor. The motion of a pendulum rocks this device so that it catches and then releases each tooth of the **escape wheel**, in turn allowing it to turn a precise amount. Unlike the **original** form used in early pendulum clocks, the anchor escapement **permitted** the pendulum to travel in a very small arc. Moreover, this invention allowed the use of a long pendulum which could beat once a second and thus led to the development of a new floor standing case design, which became known as the grandfather clock.

H

Today, highly accurate timekeeping instruments set the beat for most electronic devices. Nearly all computers contain **a quartz-crystal clock** to regulate their operation. Moreover, not only do time signals **beamed** down from Global Positioning System satellites **calibrate** the functions of precision navigation equipment, they do so as well for mobile phones, instant stock-trading systems and nationwide power-distribution grids. So **integral** have these time-based technologies become to day-to-day existence that our **dependency** on them is recognised only when they fail to work.

the anchor escapement = a type of escapement used in pendulum clocks ði

'æŋkə ɪ 'skeɪpmənt

escape wheel = a toothed wheel in the escapement of a watch or clock. ɪ 'skeɪp 'wi:l

precise = exact, correct, accurate. prɪ 'saɪs

original = initial, earliest (existing or happening first). ə 'rɪdʒɪnəl

permit = allow, enable, facilitate. pə 'mɪt

accurate = correct, precise, exact. 'ækjərət

a quartz-crystal clock = is a clock that uses an electronic oscillator that is regulated by a quartz crystal to keep time. ə kwɔ:ts- 'krɪstl 'klok

beam down = to transport somebody to or from a spaceship using special electronic equipment. bi:m daʊn

calibrate = standardize, adjust, regulate. 'kælibreɪt

precision = accuracy, exactness, correctness.

prɪ 'sɪʒn

navigation = routing, direction-finding the science or job of planning which way you need to go when you are travelling from one place to another nəvɪ 'geɪʃn

integral = connected, central, internal, forming a necessary part of something. 'ɪntɪgrəl

dependency = reliance, enslavement, craving.

dɪ 'pendənsi

READING PASSAGE 2

Air traffic control in the USA

An accident that occurred in the skies over the

Grand Canyon in 1956 resulted in the **establishment** of the **Federal Aviation Administration (FAA)** to regulate and oversee the operation of aircraft in the skies over the United States, which were becoming quite **congested**. The resulting structure of air traffic control has greatly increased the safety of flight in the United States, and similar air traffic control **procedures** are also in place over much of the rest of the wor

B

Rudimentary air traffic control (ATC) existed well before the Grand Canyon disaster. As early as the 1920s, the earliest air traffic controllers **manually** guided aircraft in the **vicinity** of the airports, using lights and flags, while **beacons** and flashing lights were placed along cross-country routes to establish the earliest airways. However, this **purely** visual system was useless in bad weather, and, by the 1930s, radio communication was

establishment = founding, launch, creation.
i'stæblɪsmənt

federal Aviation Administration (FAA) of the United States = a national authority with powers to regulate all aspects of flying in aircraft. 'fedərəl ,eɪvi'eɪʃn əd,mɪnɪ'streɪʃn (FAA)

congested = full of traffic, overfilled, blocked, crowded, #empty, #clear kən'dʒestɪd

procedure = process, way, method.
prə'si:dʒə

rudimentary = basic, elementary, simple, fundamental # advanced ru:di'mentri

manually = by hand, physically, # mental 'mænjʊəli

vicinity (of something) = neighborhood, locality, surrounding area vɪ'sɪnɪti

beacon = signal, sign, warning light, 'bi:kən
purely = entirely, wholly, totally, completely, # partly 'pjʊəli

coming into use for ATC. The first region to have something approximating today's ATC was New York City, with other major metropolitan areas following soon after.

C

In the 1940s, ATC centres could and did take advantage of the newly developed radar and improved radio communication brought about by the Second World War, but the system remained rudimentary. It was only after the creation of the FAA that full-scale **regulation** of America's airspace took place, and this was **fortuitous**, for the **advent** of the **jet engine** suddenly resulted in a large number of very fast planes, reducing pilots' margin of error and practically demanding some set of rules to keep everyone well separated and operating safely in the air.

D

Many people think that ATC **consists of** a row of controllers sitting in front of their radar screens at the nation's airports, telling arriving and departing traffic what to do. This is a very incomplete part of the picture. The FAA **realised** that the airspace over the United States would at any time have many different kinds of planes, flying for many different purposes, in a variety of weather conditions, and the same kind of structure was needed to **accommodate** all of them.

E

To meet this challenge, the following elements were **put into effect**. First, ATC extends over virtually the entire United States. In general, from 365m above the ground and higher, the entire country is **blanketed** by controlled airspace. In certain areas, mainly near airports, controlled airspace extends down to 215m above the ground, and, in the immediate vicinity of an airport, all the way down to the surface. Controlled airspace is that airspace in which FAA **regulations** apply. Elsewhere, in uncontrolled airspace, pilots are **bound** by fewer regulations. In this way, the **recreational** pilot who simply wishes to go flying for a while without all the restrictions **imposed** by the FAA has only to stay in uncontrolled airspace, below 365m, while the pilot who does want the protection **afforded** by ATC can easily enter the controlled airspace.

F

The FAA then recognised two types of operating environments. In good **meteorological** conditions, flying

metropolitan = urban, municipal, civic
metrə'pɒlɪtən

regulation = control, guideline, adjustment, rule. regjʊ'leɪʃn

fortuitous = lucky, fortunate, miraculous.
fɔ:'tju:ɪtəs

advent = arrival, beginning, initiation, # departure 'ædvent

jet engine = an engine that pushes out a stream of hot air and gases behind it, used in aircraft 'dʒet 'endʒɪn

margin of error = the degree to which a calculation might or can be wrong 'mɑ:dʒɪn əv 'erə

consist of = comprise, be made up of, be composed of, -comprise, make up. kən'sɪst ɒv

realise = recognize, understand, comprehend, # misunderstand 'ri:əlaɪz

accommodate = adapt, acclimatize, adjust. ə'kɒmədeɪt

put into effect = to make a plan or idea happen 'put 'ɪntə ɪ'fekt

virtually = almost, nearly, near. 'vɜ:tʃʊəli

blanket = to cover something with a thick layer. 'blæŋkt

regulation = rule, guideline, directive. regjʊ'leɪʃn

bind = require, force, oblige. baɪnd

recreation = fun, enjoyment, pleasure, good/great time, a blast, entertainment, relaxation, leisure. rekri'eɪʃn

impose = force, require, obey, make rules. ɪm'pəʊz

afford = give, offer, provide, allow. ə'fɔ:d

meteorological = atmospheric, climatic, weather. mi:tjərə'lɒdʒɪkl

would be permitted under Visual Flight Rules (VFR), which suggests a strong **reliance** on visual cues to maintain an acceptable level of safety. Poor visibility **necessitated** a set of Instrumental Flight Rules (IFR), under which the pilot relied on **altitude** and **navigational** information provided by the plane's instrument panel to fly safely. On a clear day, a pilot in controlled airspace can choose a VFR or IFR flight plan, and the FAA regulations were **devised** in a way which **accommodates** both VFR and IFR operations in the same airspace. However, a pilot can only choose to fly IFR if they **possess** an instrument rating which is above and beyond the basic pilot's license that must also be held.

G

Controlled airspace is divided into several different types, **designated** by letters of the alphabet. Uncontrolled airspace is designated Class F, while controlled airspace below 5,490m above sea level and not in the vicinity of an airport is Class E. All airspace above 5,490m is designated Class A. The reason for the division of Class E and Class A airspace **stems from** the type of planes operating in them. Generally, Class E airspace is where one finds general aviation aircraft (few of which can climb above 5,490m anyway), and commercial **turboprop** aircraft. Above 5,490m is the **realm** of the heavy jets, since jet engines operate more efficiently at higher altitudes. The difference between Class E and A airspace is that in Class A, all operations are IFR, and pilots must be instrument-rated, that is, skilled and licensed in aircraft **instrumentation**. This is because ATC control of the entire space is essential. Three other types of airspace, Classes D, C and B, govern the vicinity of airports. These **correspond** roughly to small **municipal**, medium-sized metropolitan and major metropolitan airports respectively, and **encompass** an increasingly **rigorous** set of regulations. For example, all a VFR pilot has to do to enter Class C airspace is establish two-way radio contact with ATC. No **explicit** permission from ATC to enter is needed, although the pilot must continue to obey all regulations **governing** VFR flight. To enter Class B airspace, such as on approach to a major metropolitan airport, an explicit ATC clearance is required. The private pilot who **cruises** without permission into this airspace risks losing their **license**.

reliance = dependence, rely on, hinge on.

ri'laɪəns

necessitate = essential, require, need, demand. ni'sesɪteɪt

cue = signal, indication, clue. kju:

altitude = height above sea level. 'æltɪtju:d

navigation = direction-finding, steering, routing. nævi'geɪʃn

devise = plan, develop, create, set up. dɪ'vaɪz

possess = own, have, hold, keep, #lack pə'zes

designate = elect, label, entitle, define. 'deziɡneɪt

stem from = arise from, originate from, come from. stem from

turboprop = an aircraft that gets power from this type of engine. tɜ:bəʊ'prop

realm = area, space, range, field. relm

instrumentation = the set of instruments used to help in controlling a machine instrumen'teɪʃn

correspond = relate, tally, link, match up. kɒrɪ'spɒnd

municipal = civic, public, community, #private mju:'nɪsɪpl

encompass = include, cover, contain, #exclude ɪn'kʌmpəs

rigorous = precise, careful, accurate 'rɪɡərəs

explicit = clear, precise, exact, #implicit. ɪk'splɪt

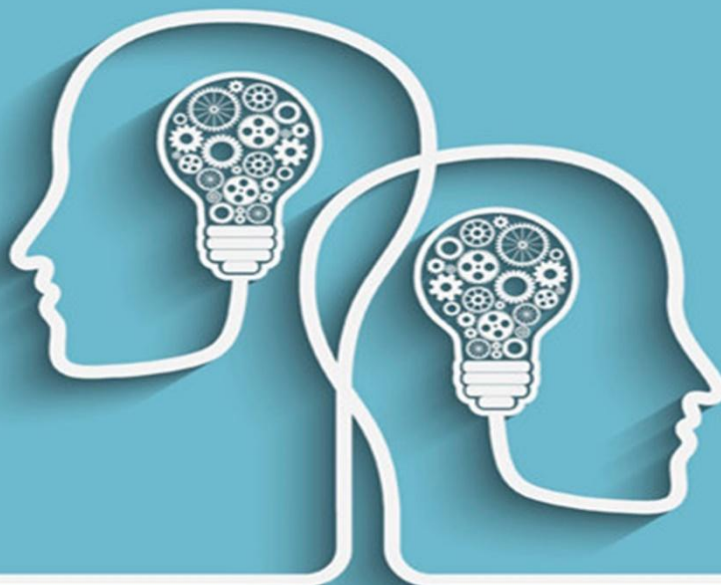
govern = rule, oversee, manage, control, regulate. 'ɡʌvɪn

cruise = fly, travel, take off, voyage. kru:z

license = certificate, pass, card, permit. 'laɪsɪns

READING PASSAGE 3

Telepathy



Can human beings communicate by thought alone?

For more than a century the issue of telepathy has divided the scientific community, and even today it still **sparks** bitter controversy among top **academics**.

Since the 1970s, **parapsychologists** at leading universities and research institutes around the world have risked the **derision** of **sceptical** colleagues by putting the various claims for telepathy to the test in dozens of rigorous scientific studies. The results and their implications are dividing even the researchers who **uncovered** them.

Some researchers say the results constitute

telepathy = mind-reading, thought transference, extrasensory perception
tɪˈlepəθi

spark = provoke, cause, trigger. spɑ:k
controversy = argument, disagreement, debate, public discussion. ˈkɒntɹəvɜːsi
academic = a teacher in a college or university. ækəˈdemɪk

parapsychology = the scientific study of mysterious abilities that some people claim to have, such as knowing what will happen in the future. pærəsaɪˈkɒlədʒi

derision = laughter, ridicule, contempt. dɪˈrɪʒən
sceptical = doubtful, untruthful, suspicious. ˈskeptɪkl

implication = suggestion, insinuation, association. ɪmˈpliːkeɪʃn

uncover = discover, reveal, expose ʌnˈkʌvə

constitute = make up, establish, create.
ˈkɒnstɪtjuːt

compelling evidence that telepathy is genuine. Other parapsychologists believe the field is on **the brink of collapse**, having tried to produce **definitive** scientific proof and failed. **Sceptics** and **advocates** alike do **concur** on one issue, however: that the most impressive evidence so far has come from the so-called 'ganzfeld' experiments, a German term that means 'whole field'. Reports of telepathic experiences had by people during **meditation** led parapsychologists to **suspect** that telepathy might involve 'signals' passing between people that were so **faint** that they were usually **swamped** by normal brain activity. In this case, such signals might be more easily detected by those experiencing meditation-like **tranquility** in a relaxing 'whole field' of light, sound and warmth.

The ganzfeld experiment tries to recreate these conditions with participants sitting in soft **reclining** chairs in a **sealed** room, listening to relaxing sounds while their eyes are covered with special filters letting in only soft pink light. In early ganzfeld **experiments**, the telepathy test involved **identification** of a picture chosen from a random selection of four taken from a large image bank. The idea was that a person acting as a 'sender' would **attempt** to **beam** the image over to the 'receiver' relaxing in the sealed room.

Once the session was over, this person was asked to identify which of the four images had been used. Random guessing would give a hit-rate of 25 per cent; if telepathy is real, however, the hit-rate would be higher. In 1982, the results from the first ganzfeld studies were **analysed** by one of its **pioneers**, the American parapsychologist Charles Honorton. They pointed to **typical** hit-rates of better than 30 per cent - a small effect, but one which **statistical** tests suggested could not be put down to chance.

compelling = forceful, convincing, persuasive, very interesting and exciting.
kəm'peɪlɪŋ

the brink of something = a situation when you are almost in a new situation, usually a bad one
ðə brɪŋk əv 'sʌmθɪŋ

collapse = fail, end, break down. kə'læps

definitive = ultimate, perfect, best. dɪ'fɪnətɪv

sceptic = cynic, doubter, questioner
#believer. 'skeptɪk

advocate = supporter, promoter, believer.
'ædvəkeɪt

concur = agree, correspond, coincide
#conflict kən'kʊ:

meditation = the practice of emptying your mind of thoughts and feelings, in order to relax completely or for religious reasons.
medɪ'teɪʃn

suspect = doubt, distrust, disbelieve. sə'spekt

faint = pale, unclear, weak #strong feɪnt

swamp = overwhelm, inundate, drown swɒmp

tranquility = calm, quiet, silence, #bustle.
træn'kwɪlɪti

reclining = rest, lie down, lounge, #stand
rɪ'klaɪnɪŋ

sealed = closed, to formally approve an agreement. si:ld

experiment = test, trial, research
ɪk'sperɪmənt

identification = recognition, classification. distinguishing
aɪ,dentɪfɪ'keɪʃn

attempt = try, make an effort, have a shot. ə'tempt

beam = send out, radiate, emit bi:m

analyze = examine, scrutinize, investigate.
'æneɪlaɪz

pioneer = creator, discoverer, inventor, forerunner paɪə'niə

typical = usual, normal, standard, average.
'tɪpɪkəl

statistical = numerical, arithmetic, arithmetical. stə'tɪstɪkəl

The implication was that the ganzfeld method had revealed real evidence for telepathy. But there was a crucial **flaw** in this argument - one routinely **overlooked** in more **conventional** areas of science. Just because chance had been ruled out as an explanation did not **prove** telepathy must exist; there were many other ways of getting positive results. These ranged from '**sensory leakage**' - where clues about the pictures accidentally reach the receiver - to **outright fraud**. In response, the researchers issued a review of all the ganzfeld studies done up to 1985 to show that 80 per cent had found statistically significant evidence. However, they also agreed that there were still too many problems in the experiments which could lead to positive results, and they drew up a list demanding new standards for future research.

After this, many researchers switched to autoganzfeld tests - an automated variant of the technique which used computers to perform many of the key tasks such as the random selection of images. By minimising human **involvement**, the idea was to minimise the risk of flawed results. In 1987, results from hundreds of autoganzfeld tests were studied by Honorton in a 'meta-analysis', a statistical technique for finding the overall results from a set of studies. Though less compelling than before, the outcome was still **impressive**. Yet some parapsychologists remain **disturbed** by the lack of **consistency** between individual ganzfeld studies. **Defenders** of telepathy point out that demanding impressive evidence from every study ignores one basic statistical fact: it takes large samples to **detect** small effects. If, as current results suggest, telepathy produces hit-rates only **marginally** above the 25 per cent expected by chance, it's unlikely to be detected by a typical ganzfeld study involving around 40 people: the group is just not big enough. Only when many studies are combined in a meta-analysis will the faint signal of telepathy really become **apparent**. And that is what researchers do seem to be finding.

What they are certainly not finding, however, is any change in attitude of **mainstream** scientists: most still totally reject the very idea of telepathy. The problem

flaw= fault, error, mistake. flo:

overlook= fail to notice, fail to see, miss.

əʊvə'lu:k

conventional =traditional, usual, conservative. kən'venʃnəl

prove = show, confirm, demonstrate. pru:v

sensory = relating to the feelings of your body rather than your mind. 'sensəri

leakage = escape, outflow, drip. 'li:kɪdʒ

outright = clear and direct, absolute, complete. 'aʊtraɪt

fraud= dishonesty, scam, deception. frɔ:d

involvement = participation, connection, contribution. ɪn'vɒlvmənt

impressive = imposing, inspiring, striking. ɪm'presɪv

disturb = perturb, concern, worry, bother. dɪ'stɜ:b

consistency= constancy, steadiness, stability. kən'sɪstənsi

defender = protector, supporter, guard. dɪ'fendə

marginally = slightly, just over, a bit 'mɑ:dʒɪnəli

detect= discover, find out, reveal, notice. dɪ'tekt

apparent = obvious, clear, seeming. ə'pærənt

mainstream = normal, typical, conventional, # unconventional 'meɪnstri:m

stems at least in part from the lack of any **plausible mechanism** for telepathy.

Various theories have been **put forward**, many focusing on **esoteric** ideas from theoretical physics. They include '**quantum entanglement**', in which events affecting one group of **atoms** instantly affect another group, no matter how far apart they may be. While physicists have demonstrated **entanglement** with specially prepared **atoms**, no-one knows if it also exists between atoms making up human minds. Answering such questions would transform parapsychology. This has **prompted** some researchers to argue that the future lies not in collecting more evidence for telepathy, but in **probing** possible mechanisms. Some work has begun already, with researchers trying to identify people who are particularly successful in autoganzfeld **trials**. Early results show that creative and artistic people do much better than average: in one study at the University of Edinburgh, musicians achieved a hit-rate of 56 per cent. Perhaps more tests like these will eventually give the researchers the evidence they are seeking and strengthen the case for the existence of telepathy.

plausible = reasonable, possible, believable.

'plɔːzəbəl

put forward = state, suggest, propose 'put
'fɔːwəd

esoteric = obscure, mysterious, cryptic, (known and understood by only a few people who have special knowledge about something). esəʊ'terɪk

quantum = a unit of energy in nuclear physics. 'kwɒntəm

entanglement = a difficult situation or relationship that is hard to escape from.

ɪn'tæŋɡlment

atom = the smallest part of an element that can exist alone or can combine with other substances to form a molecule. 'ætəm

prompt = stimulate, provoke, motivate prompt

probing = inquisitive, analytical, penetrating.

'prəʊbɪŋ

trial = test, experiment, examination. 'traɪəl

Nếu học được một lượng từ vựng lớn thì các bạn sẽ không phải quan tâm đến tip này hay trick kia khi làm bài thi IELTS Reading. Mình tin là có những bạn 1 tuần đọc liên tục được 2 cuốn Boost your vocabulary, thậm chí là hơn. Truyện dài mấy trăm trang mà nhiều bạn có thể đọc xong trong 1 đêm, còn 1 cuốn Boost your vocabulary là khá mỏng, và lại toàn từ đã được tra sẵn. Vậy nên hãy cố gắng đọc thật nhanh nhé các bạn 😊

Đinh Thắng

TEST 2

READING PASSAGE 1

Sheet glass manufacture: the float process

Glass, which has been made since the time of the Mesopotamians and Egyptians, is little more than a **mixture** of sand, soda ash and lime. When heated to about 1500 degrees Celsius (°C) this becomes a **molten** mass that **hardens** when slowly cooled. The first successful method for making clear, flat glass **involved** spinning. This method was very effective as the glass had not touched any surfaces between being soft and becoming hard, so it stayed perfectly **unblemished**, with a 'fire finish'. However, the process took a long time and was **labour intensive**.

mixture = combination, blend, hybrid, amalgam. 'mɪkstʃə

molten = metal or rock has been made into a liquid by being heated to a very high temperature. 'məʊltən

harden = solidify, freeze, consolidate, #soften 'hɑ:dən

involve = associate, engage, connect, link. ɪn'vɒlv
unblemished = flawless, perfect, untarnished, #flawed, #imperfect ʌn'blemɪʃt

labour = work, employment, hard work, manual labor 'leɪbə

intensive = concentrated, rigorous, thorough, exhaustive, #easy (tens=strain, stretch .i.e **tension**, **extension**) ɪn'tensɪv

Nevertheless, demand for flat glass was very high and glassmakers across the world were looking for a method of making it **continuously**. The first continuous **ribbon** process involved squeezing molten glass through two hot **rollers**, similar to an old **mangle**. This allowed glass of **virtually** any thickness to be made **non-stop**, but the rollers would leave both sides of the glass **marked**, and these would then need to be ground and **polished**. This part of the process **rubbed away** around 20 per cent of the glass, and the machines were very expensive.

The **float** process for making flat glass was invented by Alistair Pilkington. This process allows the **manufacture** of clear, **tinted** and **coated** glass for buildings, and clear and tinted glass for vehicles. Pilkington had been experimenting with improving the melting process, and in 1952 he had the idea of using a bed of molten metal to form the flat glass, **eliminating** altogether the need for rollers within the **float** bath. The metal had to **melt** at a temperature less than the hardening point of glass (about 600°C), but could not boil at a temperature below the temperature of the molten glass (about 1500°C). The best metal for the job was **tin**.

The rest of the **concept** **relied on gravity**, which **guaranteed** that the surface of the molten metal was perfectly flat and **horizontal**. Consequently, when **pouring** molten glass onto the molten tin, the underside of the glass would also be perfectly flat. If the glass were kept hot enough, it would flow over the molten tin until the top surface was also flat, **horizontal** and perfectly **parallel** to the bottom surface. Once the glass cooled to 604°C or less it was too hard to mark and could be transported out of the cooling zone by rollers. The glass settled to a thickness of six millimetres because of surface **tension** interactions between the glass and the tin. By **fortunate coincidence**, 60 per cent of the flat glass market at that time was for six-millimetre glass. Pilkington built a pilot plant in 1953 and by 1955 he had **convinced** his company to build a **full-scale**

continuous = uninterruptedly, endlessly, non-stop, #intermittently kən'tɪnjuəs
ribbon = length, stretch, strip 'rɪbən
roller = a piece of wood, metal or plastic, shaped like a tube, that rolls over and over. 'rəʊlə
mangle = a machine used in former times to remove water from washed clothes by pressing them between two rollers 'mæŋɡl
virtually = almost, nearly, practically. 'vɜ:tʃʊəli
non-stop = continuously, constantly, endlessly. 'nɒn'stɒp
polished = shined, cleaned, rubbed, sparkled, #tarnished. 'pɒlɪʃt
rub away = erode, wipe out, wear away 'rʌb ə'weɪ

manufacture = production, creation, making.
 mænju'fæktʃə

tinted = coloured, painted, decorated. 'tɪntɪd

coated = covered, layered, encrusted. 'kəʊtɪd

eliminate = get rid of, remove, eradicate, reject, #retain ɪ'lɪmɪneɪt

float = the surface of a liquid fləʊt

tin = a soft silver-white metal that is often used to cover and protect iron and steel 'tɪn

concept = idea, perception, belief 'kɒnsept

rely on = depend on, count on, trust rɪ'laɪ ɒn

gravity = the force that causes something to fall to the ground or to be attracted to another planet
 grævɪti

guarantee = ensure, assure. gæ'ren'ti:

pour = drizzle, tip, spill, splash. pɔ:

horizontal = flat, smooth, straight hɒrɪ'zɒntl

parallel = two lines, paths etc that are parallel to each other are the same distance apart along their whole length 'pærəlel

tension = stress pressure, strain. 'tenʃn

fortunate = lucky, happy, chance. 'fɔ:tʃənət

coincidence = when two things happen at the same time kəʊ'ɪnsɪdəns

convince = persuade, encourage, influence.
 kən'vɪns

full-scale = full-sized, complete, #partial fʊl'skeɪl

plant. However, it took 14 months of non-stop production, costing the company £100,000 a month, before the plant produced any usable glass. Furthermore, once they succeeded in making **marketable** flat glass, the machine was turned off for a service to prepare it for years of continuous production. When it started up again it took another four months to get the process right again. They finally succeeded in 1959 and there are now float plants all over the world, with each able to produce around 1000 tons of glass every day, non-stop for around 15 years.

Float plants today make glass of near **optical** quality. Several processes - melting, **refining**, **homogenising** - take place **simultaneously** in the 2000 tonnes of molten glass in the **furnace**. They **occur** in **separate** zones in a **complex** glass flow driven by high temperatures. It adds up to a continuous melting process, **lasting** as long as 50 hours, that **delivers** glass smoothly and continuously to the float bath, and from there to a coating zone and finally a heat treatment zone, where stresses formed during cooling are **relieved**.

The principle of float glass is unchanged since the 1950s. However, the product has changed **dramatically**, from a single thickness of 6.8 mm to a **range** from sub-millimetre to 25 mm, from a **ribbon** frequently **marred** by inclusions and bubbles to almost **optical** perfection. To **ensure** the highest quality, **inspection** takes place at every stage. Occasionally, a **bubble** is not removed during refining, a sand **grain** refuses to **melt**, a **tremor** in the tin puts **ripples** into the glass **ribbon**. Automated on-line **inspection** does two things. Firstly, it reveals process faults upstream that can be corrected. Inspection technology allows more than 100 million **measurements** a second to be made across the **ribbon**, locating **flaws** the **unaided** eye would be unable to see. Secondly, it enables computers downstream to **steer** cutters around flaws. Float glass is sold by the **square** metre, and at the final stage computers translate customer requirements into patterns of cuts designed to **minimise** waste.

plant = factory, workshop, manufacturing works.

plɑːnt

marketable= marketable goods, skills etc can be sold easily because people want them

ˈmɑːkɪtəbəl

optical = visual, ocular, photosensitive. ˈɒptɪkəl

refine = purify, filter, distill, # contaminate ɪˈfaɪn

homogenise = to change something so that its parts become similar or the same. (hom=same .i.e homogeneous, homosexual) həˈmɒdʒənaɪz

simultaneously= at the same time, concurrently, instantaneously sɪmlɪˈteɪniəsli

furnace= heater, boiler, oven. ˈfɜːnɪs

occur = happen, take place, befall əˈkɜː

deliver = transport, bring, carry, send. dɪˈlɪvə

relieved = released, eased, alleviated, reduced, mitigated ɪˈliːvd

dramatically= radically, noticeably, considerably, significantly. drəˈmætɪkəlɪ

range = variety, series, array. reɪndʒ

mar = spoil, ruin, detract from something, undermine mɑː

inspection = review, examination, assessment. ɪnˈspekʃn

grain = small piece, little bit, granule greɪn

tremor = shake, tremble, vibration ˈtremə

ripple = wave, undulation, wrinkle, #stillness ˈrɪpl

measurement= dimension, size, extent. ˈmeʒəmənt

unaided = bear, unprotected, unassisted ʌnˈeɪdɪd

flaw= defect, mistake, fault. floː

unaided= unassisted, without help. ʌnˈeɪdɪd

steer = drive, guide, direct. stre

cutter= a tool that is used for cutting something.

ˈkʌtə

READING PASSAGE 2



This book will provide a detailed examination of the Little Ice Age and other **climatic shifts**, but, before I **embark on** that, let me provide a historical context. We **tend** to think of climate - **as opposed to** weather - as something unchanging, yet humanity has been **at the mercy of** climate change for its entire **existence**, with at least eight **glacial** episodes in the past 730,000 years. Our ancestors adapted to the universal but **irregular** global warming since the end of the last great Ice Age, around 10,000 years ago, with **dazzling opportunism**. They developed strategies for surviving **harsh drought** cycles, decades of heavy rainfall or **unaccustomed** cold; adopted agriculture and **stock-raising**, which **revolutionised** human life; and founded the world's first pre-industrial **civilisations** in Egypt, Mesopotamia and the Americas. But the price of sudden climate change, in **famine**, disease and suffering, was often high.

climatic = relating to the weather in a particular area. *klaɪ'mætɪk*

shift = change, alteration, modification *ʃɪft*

embark on = start, begin, get on *ɪm'bɑ:k ɒn*

oppose = versus, against, contrasted with.

(op=against .i.e. opposition) ə'pəʊz

at the mercy of = unable to do anything to protect yourself from someone or something *ət ðə 'mɜ:si ɒv*

existence = being, survival, #extinction *ɪg'zɪstəns*

glacial = icy, freezing, cold, # tropical *'gleɪʃəl*

irregular = unusual, abnormal, #proper (*regul*= rule .i.e *regular, regulation*) *ɪ'regjələ*

dazzling = bright, strong, brilliant, harsh. *'dæzɪŋ*

opportunism = using every opportunity to gain power, money, or unfair advantages – used to show disapproval.

ɒpə'tju:nɪzəm

unaccustomed = unfamiliar, unusual, different, strange. *ʌnə'kʌstəmd*

stock-raising = to look after animals *'stɒk'reɪzɪŋ*

civilisation = a society that is well organized and developed, used especially about a particular place or particular time (*civ*=citizen .i.e *civic, civilian*) *sɪvəl-aɪ 'zeɪʃən*

famine = scarcity, food crisis, food shortage. *'fæmɪn*

B

The Little Ice Age lasted from roughly 1300 until the middle of the nineteenth century. Only two centuries ago, Europe experienced a cycle of bitterly cold winters; mountain **glaciers** in the Swiss Alps were the lowest in recorded memory, and pack ice surrounded Iceland for much of the year. The climatic events of the Little Ice Age did more than help shape the modern world. They are the deeply important context for the current **unprecedented** global warming. The Little Ice Age was far from a deep freeze, however; rather an **irregular** **seesaw** of rapid climatic shifts, few lasting more than a quarter-century, driven by complex and still little understood **interactions** between the atmosphere and the ocean. The seesaw brought cycles of intensely cold winters and easterly winds, then **switched abruptly** to years of heavy spring and early summer rains, **mild** winters, and frequent Atlantic storms, or to periods of droughts, light northeasterly winds, and summer **heat wave**.

C

Reconstructing the climate changes of the past is extremely difficult, because systematic weather **observations** began only a few centuries ago, in Europe and North America. Records from India and tropical Africa are even more recent. For the time before records began, we have only 'proxy records' reconstructed largely from tree rings and ice cores, **supplemented** by a few incomplete written accounts. We now have hundreds of **tree-ring** records from throughout the northern **hemisphere**, and many from south of the equator, too, **amplified** with a growing body of temperature data from ice cores **drilled** in Antarctica, Greenland, the Peruvian Andes, and other locations. We are close to a knowledge of annual summer and winter temperature **variations** over much of the northern **hemisphere** going back 600 years.

D

This book is a **narrative** history of climatic shifts during the past ten centuries, and some of the ways in which people in Europe **adapted** to them. Part One describes the Medieval Warm Period, roughly 900 to 1200. During these three centuries, **Norse voyagers** from Northern Europe explored northern seas, **settled** Greenland, and visited North America. It was not a time of uniform warmth, for then, as always since the Great Ice Age, there were constant shifts in rainfall and temperature. Mean European temperatures were about the same as today, perhaps slightly cooler.

glacier = a large mass of ice which moves slowly down a mountain valley 'glæsiə

unprecedented = extraordinary, first-time exceptional, unusual, #ordinary ʌn'presɪdɪntɪd

seesaw = alternation, oscillation, swing. 'si:so:

irregular = random, erratic, variable #regular ɪ'regjələ

interaction = communication, contact, interface. ɪntə'ræksjən

switch = change, shift, adjustment. swɪtʃ

abruptly = suddenly and unexpectedly (rupt=break .i.e **disrupt**, **interrupt**) ə'brʌptli

mild = slight, minor, weak, warm maɪld

heat wave = a period of unusually hot weather, especially one that continues for a long time, #cold spell hi:t weɪv

reconstruct = rebuilding, recreate, modernize (struct= build .i.e **construction**, **structure**) ri:kən'strʌkt

observation = surveillance, scrutiny, watching, #neglect ɒbzə'veɪʃən

proxy = substitution, deputation, delegation 'prɒksi

supplement = addition, extra, complement. (ple=fill, full .i.e **replete**, **plethora**) 'sʌplɪmənt

tree-ring = one of the rings that you can see in a tree trunk (= centre part) if you cut through it. tri:- rɪŋ

hemisphere = a half of the Earth, especially one of the halves above and below the equator (hemi=half.i.e **hemicycle**, **hemicube**) 'hemɪsfɪə

amplify = increase, strengthen, #reduce 'æmplɪfaɪ

drill = pierce, penetrate, make a hole dɹɪl

variation = difference, distinction, #similarity veəri'eɪʃən

narrative = story, tale, description 'nærətɪv

adapt = familiarize, get used to, adjust. ə'dæpt

norse = relating to the people of ancient Scandinavia or their language. nɔ:s

voyager = traveler, explorer, adventurer 'vɔɪdʒə

settle = stay, set up house, inhabit 'setl

uniform = unchanging, constant, unvarying, #uneven 'ju:nɪfɔ:m

E

It is known that the Little Ice Age cooling began in Greenland and the Arctic in about 1200. As the Arctic ice pack spread southward, Norse voyages to the west were **rerouted** into the open Atlantic, then ended altogether. Storminess increased in the North Atlantic and North Sea. Colder, much wetter weather **descended** on Europe between 1315 and 1319, when thousands **perished** in a **continent-wide famine**. By 1400, the weather had become decidedly more **unpredictable** and stormier, with sudden **shifts** and lower temperatures that **culminated** in the cold decades of the late sixteenth century. Fish were a **vital commodity** in growing towns and cities, where food supplies were a **constant** concern. Dried **cod** and **herring** were already the **staples** of the European fish trade, but changes in water temperatures forced fishing fleets to work **further offshore**. The Basques, Dutch, and English developed the first offshore fishing boats adapted to a colder and stormier Atlantic. A gradual agricultural **revolution** in Northern Europe **stemmed from** concerns over food supplies at a time of rising populations. The revolution involved **intensive commercial** farming and the growing of animal **fodder** on land not **previously** used for crops. The increased productivity from farmland made some countries **self-sufficient** in grain and livestock and offered effective protection against famine.

F

Global temperatures began to rise slowly after 1850, with the beginning of the Modern Warm Period. There was a **vast migration** from Europe by land-hungry farmers and others, to which the famine caused by the Irish potato **blight** contributed to North America, Australia, New Zealand, and southern Africa. Millions of hectares of forest and woodland fell before the newcomers' axes between 1850 and 1890, as **intensive** European farming methods expanded across the world. The **unprecedented** land **clearance** released vast quantities of carbon dioxide into the atmosphere, **triggering** for the first time humanly caused global warming. Temperatures climbed more rapidly in the twentieth century as the use of **fossil** fuels **proliferated** and greenhouse gas levels continued to **soar**. The rise has been even **steeper** since the early 1980s. The Little Ice Age has given way to a new climatic **regime**, **marked** by **prolonged** and steady warming. At the same time, extreme weather events like Category 5 **hurricanes** are becoming more frequent.

reroute = redirect, deflect, switch ri'ru:t

descend = fall down, fall, decline, #ascend (de=decline .i.e decrease, decline, destroy) di'send

perish = die, pass away, decease, #live, #survive 'periʃ

continent = mainland, landmass, landform, land 'kontinənt

culminate = end, finish, #start. 'kʌlmɪneɪt

cod = a large sea fish that lives in the North Atlantic kɒd

herring = a long thin silver sea fish that can be eaten. 'herɪŋ

staple = a food that is needed and used all the time 'steɪpl

offshore = in or under the sea and not far from the coast. ɒf'ʃɔ:

stem from = arise from, come from, be a result of. stem from

fodder = food, silage, rations, feed 'fɒdə

self-sufficient = independent, autonomous, self-supporting self sə'fɪʃnt

vast = huge, massive, enormous. vɔ:t

migration = relocation, movement, immigration, resettlement maɪ'greɪʃn

blight = disease, an unhealthy condition of plants in which parts of them dry up and die. blaɪt

unprecedented = unusual, exceptional, rare. ʌn'presɪdəntɪd

clearance = permission, authorization, allowance. 'klɪərəns

trigger = activate, cause, elicit, #halt. 'trɪgə

proliferate = increase, multiply, grow. prə'lɪfəreɪt

soar = increase, rise, escalate, #plummet sə:

steep = sheer, sharp, vertical sti:p.

regime = system, establishment. rei'ʒɪm

prolonged = continued, extended, long, sustained, # brief, short-lived prə'lɒŋd

hurricane = storm, cyclone, typhoon, tornado 'hʌrɪkən

READING PASSAGE 3

The meaning and power of smell

The sense of smell, or **olfaction**, is powerful.

Odours affect us on a physical, psychological and social level. For the most part, however, we breathe in the **aromas** which surround us without being **consciously aware** of their importance to us. It is only when the **faculty** of smell is **impaired** for some reason that we begin to realise the essential role the sense of smell plays in our sense of well-being

A survey conducted by Anthony Synott at Montreal's Concordia University asked participants to comment on how important smell was to them in their lives. It became apparent that smell can **evoke** strong emotional responses. A scent associated with a good experience can bring a **rush** of joy, while a **foul odour** or one associated with a bad memory may make us **grimace** with **disgust**. **Respondents** to the survey noted that many of their olfactory likes and dislikes were based on emotional associations. Such

olfaction = the action of smelling ol'fæk.jən
odour = smell, whiff, scent, fragrance, perfume, aroma. 'əʊdə
aroma = a pleasant smell, especially from food or coffee. ə'roumə
consciously = aware, intentionally, on purpose, unintentionally 'kɒnʃəsli
faculty = a natural ability, capacity, sense, #inability 'fæklti
impair = harm, damage, weaken, worsen, #enhance ɪm'peə

conduct = do, make, carry out. (duc=make .i.e produce, introduce) kən'dʌkt
evoke = induce, arouse, stir up, #suppress ɪ'vəʊk
rush = flow, pour, gush, stream rʌʃ
foul = unpleasant, disgusting, horrible. faʊl
grimace = twist, pull a face, make a face, #smile. grɪ'meɪs
disgust = revulsion, repugnance, loathing, hatred, #attraction dɪs'gʌst
respondent = responder, participant, interviewee, answerer. rɪ'spɒndənt

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Biên tập cuốn sách Boost your vocabulary này được thực hiện bởi [Đinh Thắng](#) & [IELTS Family-Các nhóm tự học IELTS](#)

associations can be powerful enough so that odours that we would generally label unpleasant become agreeable, and those that we would generally consider fragrant become disagreeable for particular individuals. The perception of smell, therefore, **consists** not only of the sensation of the odours themselves, but of the experiences and emotions associated with them.

B

Odours are also essential **cues** in social **bonding**. One respondent to the survey believed that there is no true emotional bonding without touching and smelling a loved one. In fact, **infants** recognise the odours of their mothers soon after birth and adults can often identify their children or **spouses** by scent. In one well-known test, women and men were able to **distinguish** by smell alone clothing worn by their marriage partners from similar clothing worn by other people. Most of the subjects would probably never have given much thought to odour as a cue for identifying family members before being involved in the test, but as the experiment revealed, even when not consciously considered, smells **register**.

C

In spite of its importance to our emotional and **sensory** lives, smell is probably the most **undervalued sense** in many cultures. The reason often given for the low regard in which smell is held is that, in comparison with its importance among animals, the human sense of smell is **feeble** and undeveloped. While it is true that the olfactory powers of humans are nothing like as fine as those **possessed** by certain animals, they are still **remarkably acute**. Our noses are able to recognise thousands of smells, and to perceive odours which are present only in extremely small quantities.

D

Smell, however, is a highly **elusive phenomenon**. Odours, unlike colours, for instance, cannot be named in many languages because the specific vocabulary simply doesn't exist. 'It smells like...', we have to say when describing an odour, **struggling** to express our olfactory experience. Nor can odours be recorded: there is no effective way to either capture or store them over time. In the **realm** of olfaction, we must make do with descriptions and recollections. This has **implications** for olfactory research.

association = connection, involvement, correlation əˌsəʊʃiˈeɪʃn

consist= contain, involve, comprise kənˈsɪst

sensation=feeling, sense, awareness senˈseɪʃn

cue = hint, clue, signal, sign kjuː

bonding= connection, relationship, association. ˈbɒndɪŋ

infant= baby, child, newborn. ˈɪnfənt

spouse = husband/wife, partner, other half. spauz

distinguish = recognize, identify, discern. dɪˈstɪŋɡwɪʃ

register = realize, notice= if something registers, or if you register it, you realize or notice it, and then remember it ˈredʒɪstə

sensory= sensual, bodily, #intellectual (sens=feel.i.e **sensitive**, **sensible**) ˈsensəri

undervalued = underestimate, underrated. ʌndəˈvæljuːd

feeble = weak, ineffective, poor, #strong, #effective ˈfiːbl

possess = have, hold, own, retain, #lack pəˈzes

remarkably = extraordinarily, amazingly, outstandingly, extremely. ɪˈmɑːkəbli

acute = sharp, sensitive, heightened əˈkjuːt

perceive= notice, sense, recognize. pəˈsiːv

elusive = indefinable, indescribable, hard to pin down. ɪˈluːsɪv

phenomenon = occurrence, fact, event, happening fɪˈnɒmɪnən

struggle =strive, strain, make an effort ˈstrʌɡl

capture = catch, seize, take, pick up, #release. (capt=hold, take .i.e **captivate**) ˈkæptʃə

realm = area, field, department, scope. relm

implication = suggestion, association, insinuation. ɪmˈpliˈkeɪʃn

E

Most of the research on smell **undertaken** to date has been of a physical scientific nature. Significant advances have been made in the understanding of the biological and chemical nature of olfaction, but many fundamental questions have yet to be answered.

Researchers have still to decide whether smell is one sense or two - one responding to odours **proper** and the other registering odourless chemicals in the air. Other unanswered questions are whether the nose is the only part of the body affected by odours, and how smells can be measured **objectively** given the nonphysical components. Questions like these mean that interest in the **psychology** of smell is **inevitably** set to play an increasingly important role for researchers.

F

However, smell is not simply a biological and psychological phenomenon. Smell is cultural, hence it is a social and historical phenomenon. Odours are **invested** with cultural values: smells that are considered to be **offensive** in some cultures may be perfectly acceptable in others. Therefore, our sense of smell is a means of, and **model** for, interacting with the world. Different smells can provide us with intimate and emotionally charged experiences and the value that we attach to these experiences is **interiorised** by the members of society in a deeply personal way. Importantly, our commonly held feelings about smells can help distinguish us from other cultures. The study of the cultural history of smell is, therefore, in a very real sense, an investigation into the essence of human culture.

undertake = carry out, do # neglect ʌndə'teɪk

proper = correct, appropriate, accurate 'prɒpə

objectively = accurately, empirically, demonstrably, tangibly, #subjectively əb'dʒektɪvli

psychology = the mental processes involved in believing in something or doing a certain activity saɪ'kɒlədʒi

inevitably = predictably, unsurprisingly, without doubt. ɪn'evɪtəbli

invest = supply, enable, put in. ɪn'vest

offensive = unpleasant, distasteful, disgusting ə'fensɪv

model = example, type, sort, genre 'mɒdl

intimate = private, personal, secret, #public 'ɪntɪmeɪt

attach = connect, stick, glue, #detach ə'tætʃ

interior = inner, inside >< exterior. ɪn'tɪəriə

essence = the most basic and important quality of something. 'esns

Nhiều bạn ngại là đọc cuốn này xong thì làm test không đánh giá đúng nữa? Thật ra câu trả lời là **KHÔNG PHẢI NHƯ VẬY**. Ở đầu sách đã ghi rất rõ là bạn cần phải làm test trước và sau đó thì dùng cuốn này để không phải mất công tra từ điển, cộng thêm với việc học synonym (từ đồng nghĩa) để hiểu đoạn văn nói gì.

Tuy nhiên, với các bạn band Reading đã ở tầm 7.0-8.0 thì cũng không cần phải làm test quá nhiều nữa. Tập trung vào đọc bài cho thật hiểu, đến từ nào không biết thì nhìn sang cột bên phải quyển Boost này để xem nghĩa của từ và lại đọc tiếp. **ĐỌC, ĐỌC, ĐỌC. HIỂU, HIỂU, HIỂU.** Cứ thế đọc mấy cuốn này như đọc báo, **KHÔNG PHẢI LÀM TEST NHIỀU**. Tự khắc điểm sẽ lên 8.0-9.0. Vì nếu học từ vựng mà không hiểu nội dung bài đọc thật sâu thì cũng vô nghĩa.

Hy vọng là sách bộ Boost your vocabulary - Cambridge IELTS này sẽ tiếp tục giúp được nhiều bạn tiết kiệm thời gian & đạt kết quả thật cao trong Reading!

Đình Thắng

TEST 3

READING PASSAGE 1

Striking Back at Lightning With Lasers

Seldom is the weather more dramatic than when thunderstorms **strike**. Their electrical **fury** **inflicts** death or serious **injury** on around 500 people each year in the United States alone. As the clouds **roll** in, a **leisurely** **round** of golf can become a terrifying **dice with death** - **out in the open**, a lone golfer may be a **lightning bolt's** most inviting target. And there is damage to **property** too. Lightning damage costs American power companies more than \$100 million a year.

But researchers in the United States and Japan are planning to hit back. Already in **laboratory** trials they have tested strategies for **neutralising** the power of thunderstorms, and this winter they will **brave** real

seldom = rarely, infrequently, occasionally. 'seldəm
strike = hit, attack, crash into. straɪk
fury = extreme anger (often uncontrolled anger), rage, violence. 'fjʊəri
inflict = impose, cause, perpetrate. ɪn'flɪkt
leisurely = slow, unhurried, relaxed, #rushed 'leɪzəli
dice with death = to do something extremely dangerous and silly 'daɪs wɪð deθ
out in the open = apparent, clear, not hidden or secret aʊt ɪn ðɪ 'əʊpən
a lightning bolt's = a flash of lightning in the sky ə 'laɪtnɪŋ bəʊlt's

laboratory = workroom, test center, workshop (research laboratory). lə'bɒrətəri
neutralize = balance out, counteract, make safe, reduce the effect. 'nju:trəlaɪz
brave the elements/weather etc = go out in bad weather breɪv ðɪ 'elɪments/ 'weðə et'setə

storms, **equipped** with an **armoury** of lasers that they will be pointing towards the heavens to **discharge** **thunderclouds** before lightning can strike.

The idea of forcing storm clouds to discharge their lightning on **command** is not new. In the early 1960s, researchers tried **firing** rockets **trailing wires** into thunderclouds to set up an easy discharge path for the huge electric charges that these clouds **generate**. The technique survives to this day at a test site in Florida run by the University of Florida, with support from the Electrical Power Research Institute (EPRI), based in California. EPRI, which is **funded** by power companies, is looking at ways to protect the United States' power grid from lightning strikes. 'We can cause the lightning to strike where we want it to using rockets,' says Ralph Bernstein, manager of lightning projects at EPRI. The rocket site is providing precise measurements of lightning **voltages** and allowing engineers to check how electrical equipment **bears up**.

Bad behavior

But while rockets are fine for research, they cannot provide the protection from lightning strikes that everyone is looking for. The rockets cost around \$1,200 each, can only be fired at a limited **frequency** and their failure rate is about 40 per cent. And even when they do **trigger** lightning, things still do not always go **according to** plan. 'Lightning is not perfectly **well behaved**,' says Bernstein. 'Occasionally, it will take a **branch** and go someplace it **wasn't supposed to** go.'

And anyway, who would want to fire streams of rockets in a populated area? 'What goes up must come down,' points out Jean-Claude Diels of the University of New Mexico. Diels is leading a project, which is **backed** by EPRI, to try to use lasers to discharge lightning safely- and safety is a basic **requirement** since no one wants to put themselves or their expensive equipment **at risk**. With around

equip = prepare, provide, give. ɪ'kwɪp
armoury = a place where weapons are stored. 'ɑ:məri
discharge = release, send out, free. 'dɪstʃɑ:dʒ
thundercloud = a large dark cloud that you see before or during a storm 'θʌndəklaʊd

command = order, directive, charge. kə'mɑ:nd

fire = shoot, trigger, launch, set off. 'faɪə

wire = cable, line, chain 'waɪə

trailing wire = a flexible insulated cable used for transmitting power from the main power source to a mobile machine 'treɪlɪŋ 'waɪə

generate = make, produce, create. 'dʒenəreɪt

fund = sponsor, finance, support, fʌnd

voltage = power, energy, electrical energy. 'vəʊltɪdʒ

bear up = cope, survive, manage 'beər 'ʌp

frequency = regularity, incidence, occurrence, rate of recurrence 'fri:kwənsi

trigger = activate, start, set off. 'trɪɡə

according to = as said by, as stated by, in accordance with ə'ko:dnɪŋ tu:

well behaved = polite, respectful, well-mannered wel bi'heɪvd

branch = part, section, division. brɑ:ntʃ

be supposed to = should, ought to, be expected to bi sə'pəʊzd tu:

back = sponsor, support, finance, fund. 'bæk

requirement = obligation, condition, necessity

#option ɪ'kwɑɪəmənt

at risk = in danger, at stake, endangered, vulnerable, #safe ət rɪsk

\$500,000 invested so far, a **promising** system is just **emerging** from the laboratory.

The idea began some 20 years ago, when high-powered lasers were **revealing** their ability to **extract** electrons out of atoms and **create** ions. If a **laser** could **generate** a **line** of **ionisation** in the air all the way up to a storm cloud, this **conducting path** could be used to guide lightning to Earth, before the electric field becomes strong enough to break down the air in an **uncontrollable surge**. To stop the **laser** itself being struck, it would not be pointed straight at the clouds. Instead it would be directed at a mirror, and from there into the sky. The mirror would be protected by placing lightning conductors close by. Ideally, the cloud-zapper (gun) would be cheap enough to be **installed** around all key power installations, and **portable** enough to be taken to international sporting events to **beam up** at **brewing** storm clouds.

A stumbling block

However, there is still a big **stumbling block**. The **laser** is no **nifty portable**: it's a **monster** that takes up a whole room. Diels is trying to cut down the size and says that a **laser** around the size of a small table is in the **offing**. He plans to test this more manageable system on live thunderclouds next summer. Bernstein says that Diels's system is attracting lots of interest from the power companies.

But they have not yet come up with the \$5 million that EPRI says will be needed to develop a **commercial** system, by making the lasers yet smaller and cheaper. I cannot say I have money yet, but I'm working on it,' says Bernstein. He **reckons** that the **forthcoming** field tests will be the **turning point** - and he's hoping for good news. Bernstein predicts 'an **avalanche** of interest and support' if all goes well. He expects to see cloud-zappers **eventually** costing \$50,000 to \$100,000 each.

Other scientists could also benefit. With a lightning 'switch' **at their fingertips**, materials scientists could find out what happens when **mighty** currents meet matter. Diels also hopes to see the birth of 'interactive **meteorology**' - not just **forecasting** the weather but

promising = hopeful, likely, capable, favorable, #disappointing 'prɒmɪsɪŋ
emerge= appear, come out, begin. ɪ'mɜːdʒ

reveal = disclose, expose, uncover, bring to light, #cover up rɪ'viːl

extract = remove, pull out, take out. ɪk'strækt

ionise = to form ions or make them form 'aɪ.ən.aɪz

conducting path = a path that electricity can flow through kən'dʌktɪŋ pɑːθ

uncontrollable = unmanageable, wild, out of control, uncontrollable ʌnkən'trəʊləbəl

surge = rise, growth, spread, # decline sɜːdʒ

install = put in, connect, set up. ɪn'stɔːl

portable = moveable, handy, transportable. 'pɔːtəbəl

beam = to send out a line of light, heat, energy etc biːm

brewing = if a storm is brewing, it will happen soon. 'bruːɪŋ

stumbling block = obstacle, problem, difficulty, barrier 'stʌmbəlɪŋ blɒk

nifty = useful, convenient, effective, #useless 'nɪftɪ

offing (be in the offing) = be imminent, be likely, loom, be on the horizon 'ɒfɪŋ

commercial= profitable, marketable, profit-making. kə'mɜːʃl

reckon = think, calculate, suppose. 'rekən

forthcoming= approaching, upcoming, future. fɔːθ'kʌmɪŋ

turning point = decisive moment, crossroads. (the time when an important change starts, especially one that improves the situation). 'tɜːnɪŋ point

avalanche = a very large number of things. 'ævələːntʃ

at one's fingertips = convenient, handy, easy, accessible ət wʌnz 'fɪŋɡətɪps

mighty= strong, powerful, great. 'maɪti

current= flow, stream, tide. 'kʌrənt

meteorology = climatology, weather mi'tiɒ'rɒlədʒi

forecast = predict, estimate. fɔːkɑːst

controlling it. 'If we could discharge clouds, we might affect the weather,' he says.

And perhaps, says Diels, we'll be able to **confront** some other meteorological **menaces**. 'We think we could prevent **hail** by inducing lightning,' he says. Thunder, the shock wave that comes from a lightning flash, is thought to be the trigger for the torrential rain that is typical of storms. A laser thunder factory could **shake** the moisture **out of** clouds, perhaps preventing the **formation** of the giant hailstones that threaten crops. With luck, as the storm clouds gather this winter, laser-toting researchers could, for the first time, **strike back**.

confront = tackle, face, deal with. kən'frʌnt

menace = threat, danger, risk. 'menəs

hail= frozen raindrops, sleet, frozen rain, hailstones heɪl

shake sth out of sth= get rid of, remove. ʃeɪk 'sʌmθɪŋ aʊt əv 'sʌmθɪŋ

formation= creation, development, establishment. fɔ:'meɪʃn

strike back= revenge, retaliate, fight back = to attack or criticize someone who attacked or criticized you first 'straɪk 'bæk

READING PASSAGE 2

The Nature of Genius

There has always been an interest in geniuses and **prodigies**. The word 'genius', from the Latin gens (= family) and the term 'genius', meaning 'begetter', comes from the early Roman **cult** of a **divinity** as the head of the family. In its earliest form, genius was concerned with the ability of the head of the family, the **paterfamilias**, to **perpetuate** himself. **Gradually**, genius came to represent a person's characteristics and thence an individual's highest **attributes derived from** his 'genius' or guiding spirit. Today, people still look to stars or genes, **astrology** or genetics, in the hope of finding the source of **exceptional** abilities or personal characteristics.

The concept of genius and of gifts has become part of our **folk** culture, and attitudes are **ambivalent** towards them. We **envy** the gifted and **mistrust** them. In the

genius = talent, gift, flair, expertise. 'dʒiːniəs
prodigy = genius, a young person who has a great natural ability in a subject or skill. 'prɒdɪdʒi
cult = a system of religious beliefs and practices kʌlt
divinity = religion, theology, spirituality. drɪˈvɪnɪti
paterfamilias = father, headman, paternalist pətəfəˈmɪliəs
perpetuate = continue, maintain, extend, preserve. pəˈpetʃueɪt
gradually = slowly, regularly, steadily. 'grædʒʊəli
attribute = trait, feature, characteristic, quality. 'ætrɪbjʊt
derive from = originate, stem, arise drɪˈraɪv frɒm
astrology = horoscope, the signs of the zodiac, star sign/sign. əˈstrɒlədʒi
exceptional = excellent, brilliant, extraordinary, outstanding ɪk'sepʃnəl

folk = traditional, widespread, popular. fəʊk
ambivalent = unsure, hesitant, uncertain. æmˈbɪvələnt
envy = covet, be jealous of, resent, #goodwill 'envi
mistrust = distrust, doubt, disbelieve. mɪs'trʌst

mythology of giftedness, it is popularly believed that if people are talented in one area, they must be **defective**

In another, that **intellectuals** are **impractical**, that prodigies burn too brightly too soon and **burn out**, that **gifted** people are **eccentric**, that they are physical weaklings, that there's a thin line between genius and madness, that genius runs in families, that the gifted are so clever they don't need special help, that giftedness is the same as having a high IQ, that some races are more intelligent or musical or mathematical than others, that genius goes **unrecognised** and **unrewarded**, that **adversity** makes men **wise** or that people with gifts have a responsibility to use them. Language has been **enriched** with such terms as 'highbrow', 'egghead', 'blue-stocking', 'wiseacre', 'know-all', 'boffin' and, for many, 'intellectual' is a term of **denigration**.

The nineteenth century saw considerable interest in the nature of genius, and produced not a few studies of famous prodigies. Perhaps for us today, two of the most significant aspects of most of these studies of genius are the frequency with which early **encouragement** and teaching by parents and tutors had beneficial effects on the intellectual, artistic or musical development of the children but caused great difficulties of **adjustment** later in their lives, and the frequency with which abilities went unrecognised by teachers and schools. However, the difficulty with the evidence produced by these studies, **fascinating** as they are in collecting together **anecdotes** and **apparent** similarities and exceptions, is that they are not what we would today call **norm-referenced**. In other words, when, for instance, information is **collated** about early illnesses, methods of upbringing, schooling, etc., we must also **take into account** information from other historical sources about how common or exceptional these were at the time. For instance, infant **mortality** was high and **life expectancy** much shorter than today, home tutoring was common in the families of the **nobility** and wealthy, **bullying** and

defective = faulty, imperfect, unreliable, #perfect. *di'fektiv*

intellectual = philosopher, thinker, scholar. *intə'lektʃuəl*

impractical = unrealistic, unreasonable, #practical. *im'præktɪkəl*

burn out = exhaust, break down, wear out *'bɜ:n 'aʊt*

gifted = talented, exceptional, remarkable *'gɪftɪd*

eccentric = odd, strange, weird, unusual, peculiar. *ek'sentɪk*

weakling = someone who is not physically strong. *'wi:kɪŋ*

unrecognised = anonymous, unidentified, unknown *ʌn'rekəg,naɪzd*

unrewarded = unpaid, uncompensated, #paid *ʌnrɪ'wɔ:dɪd*

adversity = hardship, difficulty, hard times. *əd'vɜ:səti*

wise = intelligent, clever, bright, brilliant. *waɪz*

enrich = improve, enhance, develop, augment. *ɪn'ri:tʃ*

denigrate = disparage, degrade, #praise to say things to make someone or something seem less important or good. *'denɪgreɪt*

encouragement = reassurance, inspiration, reinforcement, #discouragement *ɪn'kʌrɪdʒmənt*

adjustment = change, alteration, modification. *əd'ʒʌstmənt*

fascinating = interesting, stimulating, intriguing, #repellent, #repellent. *'fæsɪneɪtɪŋ*

anecdote (a short story based on our personal experience) = story, tale, narration *'ænikdəʊt*

apparent = obvious, clear, evident. *ə'pærənt*

norm-referenced = reference to an accepted standard or a way of behaving or doing things that most people agree with *nɔ:m-'refrənst*

collate = collect, compare, gather. *kə'leɪt*

take into account = consider, include, bear in mind, think about, take into consideration. *'teɪk 'ɪntə ə'kaʊnt*

mortality = the number of deaths *mɔ:'tælɪti*

life expectancy = lifespan, lifetime, natural life *'laɪf ɪk'spektənsi*

nobility = upper class, superiority, cream of society. *nəʊ'bɪlɪti*

bully = persecute, oppress, harass = to threaten to hurt someone or frighten them *'buli*

corporal punishment were common at the best independent schools and, for the most part, the cases studied were members of the **privileged** classes. It was only with the growth of **paediatrics** and psychology in the twentieth century that studies could be **carried out** on a more objective, if still not always very scientific, basis.

Geniuses, however they are defined, are but the **peaks** which **stand out** through the mist of history and are visible to the particular **observer** from his or her particular **vantage point**. Change the observers and the vantage points, clear away some of the mist, and a different lot of peaks appear. Genius is a term we apply to those whom we recognise for their outstanding achievements and who stand near the end of the **continuum** of human abilities which reaches back through the **mundane** and **mediocre** to the incapable. There is still much truth in Dr Samuel Johnson's **observation**. The true genius is a mind of large general powers, accidentally **determined** to some particular direction'. We may disagree with the 'general', for we doubt if all musicians of genius could have become scientists of genius or vice versa, but there is no doubting the accidental determination which **nurtured** or **triggered** their gifts into those channels into which they have **poured** their powers so successfully. Along the continuum of abilities are hundreds of thousands of gifted men and women, boys and girls.

What we appreciate, enjoy or marvel at in the works of genius or the achievements of prodigies are the **manifestations** of skills or abilities which are similar to, but so much **superior** to, our own. But that their minds are not different from our own is demonstrated by the fact that the **hard-won** discoveries of scientists like Kepler or Einstein become the **commonplace** knowledge of schoolchildren and the once **outrageous**

corporal = physical, bodily, #spiritual, #mental 'kɔ:pərəl
privileged = rich, wealthy, affluent, prosperous, well-off. 'prɪvəlɪdʒd
paediatrics = the area of medicine that deals with children and their illnesses. pi:di'ætriks
carried out = conduct, do, perform, accomplish. 'kæriɪd 'aut

peak = top, pinnacle, apex, #bottom. pi:k
stand out = be obvious, be noticeable, be conspicuous. stænd 'aut
mist = haze, fog, smog mist
the mist of history = a period of time so long ago that people cannot remember it ðə mist əv 'hɪstri
vantage point = point of view, perspective, viewpoint. 'vɑ:ntɪdʒ pɔɪnt
continuum = range, field, scale. kən'tɪnjuəm
mundane = boring, dull, tedious, monotonous. mʌn'deɪn
mediocre = average, ordinary, middling, not very good #excellent mi:di'əʊkə
incapable = unable, incompetent, #capable ɪn'keɪpəbl
determined = strong-minded, firm, fixed dɪ'tɜ:mɪnd
vice versa = the opposite of a situation you have just described is also true. vaɪsi 'vɜ:sə
nurture = raise, foster, bring up, 'nɜ:tʃə
trigger = cause, activate, generate, #halt 'trɪɡə
pour sth into sth = to give a lot of money or effort to something with the idea of making it successful pɔ: 'sʌmθɪŋ 'ɪntə 'sʌmθɪŋ

marvel = admire, be amazed, gaze in awe 'mɑ:vəl
manifestation = appearance, display, show. mænɪfe'steɪʃn
superior = greater, higher, more. su:'prɪəriə
hard-won = achieves only after a lot of effort and difficulty. hɑ:d- wʌn
commonplace = common, widespread, ordinary. 'kɒmənpleɪs
outrageous = disgraceful, shocking, extreme, #commendable
 (extremely unusual and slightly amusing or shocking) ɔ:t'reɪdʒəs

shapes and colours of an artist like Paul Klee so soon appear on the **fabrics** we wear. This does not minimise the **supremacy** of their achievements, which outstrip our own as the sub-four-minute milers outstrip our jogging.

To think of geniuses and the gifted as having uniquely different brains is only reasonable if we accept that each human brain is uniquely different. The purpose of instruction is to make US even more different from one another, and in the process of being educated we can learn from the achievements of those more gifted than ourselves. But before we try to **emulate** geniuses or encourage our children to do so we should note that some of the things we learn from them may **prove** **unpalatable**. We may envy their achievements and **fame**, but we should also recognise the price they may have paid in terms of **perseverance**, **single-mindedness**, **dedication**, **restrictions** on their personal lives, the demands upon their energies and time, and how often they had to display great courage to preserve their integrity or to make their way to the top.

Genius and giftedness are relative **descriptive** terms of no real **substance**. We may, at best, give them some precision by defining them and placing them in a context but, whatever we do, we should never **delude** ourselves into believing that gifted children or geniuses are different from the rest of humanity, save in the degree to which they have developed the performance of their abilities.

fabric = material, cloth, textiles. 'fæbrɪk
supremacy = superiority #inferiority (the position in which you are more powerful or advanced than anyone else). suˈpreməsi
outstrip = outdo, surpass, better, do better. aʊtˈstri:p

emulate = imitate, follow, copy, mimic. 'emjʊleɪt
prove = show, evidence, verify. pruːv
unpalatable = unpleasant, disagreeable, unacceptable. ʌnˈpælətəbəl
fame = recognition, reputation, prominence feɪm
perseverance = determination to keep trying to achieve something in spite of difficulties. pɜːsɪ'veɪərəns
single-minded = determined, headstrong, persistent. sɪŋgl 'maɪndɪd
dedication = devotion, commitment, keenness. dedɪ'keɪʃn
restriction = limit, restraint, constraint. rɪ'strɪkʃn

descriptive = explanatory, illustrative, expressive. dɪ'skrɪptɪv
substance = stuff, material, matter. 'sʌbstəns
precision = exactness, accuracy, correctness. prɪ'sɪʒn
delude = deceive, cheat, mislead, pull the wool over somebody's eyes dɪ'luːd

READING PASSAGE 3



How does the biological clock tick?

Our **life span** is **restricted**. Everyone accepts this as 'biologically' obvious. 'Nothing lives for ever!' However, in this statement we think of **artificially** produced, **technical** objects, products which are subjected to natural wear and tear during use. This leads to the result that at some time or other the object stops working and is unusable ('death' in the biological **sense**). But are the wear and tear and loss of **function of technical** objects and the death of living **organisms** really **similar** or **comparable**?

Our 'dead' products are '**static**', closed systems. It is always the basic material which **constitutes** the object and which, in the natural course of things, is **worn down** and becomes 'older'. **Ageing** in this case must **occur** according to the laws of **physical** chemistry and of **thermodynamics**. Although the same law holds for a living **organism**, the result of this law is not **inexorable** in the same way. At least as long as a biological system has the ability to renew itself it could actually become older without ageing; an **organism** is an open, **dynamic** system through which new material continuously flows.

biological clock = body clock, internal clock,

biological rhythm baɪə'lədʒɪkəl 'klɒk

tick = makes a short repeated sound tɪk

life span = lifetime, life expectancy, natural life

'laɪf spæn

restrict = limit, constrain, constrict. rɪ'strɪkt

artificially = synthetic, man-made, unnaturally,

#naturally. ɑ:tɪ'fɪʃli

organism = an animal, plant, human or any

other living thing. 'ɔ:ɡənɪzəm

static = still, standing, inactive >< moving.

'stæɪtɪk

constitute = to be considered to be something

'kɒnstɪtju:t

wear down = to gradually become flatter or smoother, or to make something become like this, because of rubbing or use weə daʊn

ageing = grow older, become old, get older

'eɪdʒɪŋ

thermodynamics = the science that deals with the relationship between heat and other forms of energy. θɜ:məʊdaɪ'næmɪks

inexorable = unstoppable, inevitable, unavoidable. ɪn'eksərəbəl

dynamic = lively, active, energetic daɪ'næmɪk

Destruction of old material and formation of new material are thus in **permanent dynamic equilibrium**. The **material** of which the **organism** is formed changes continuously. Thus our bodies continuously exchange old substance for new, just like a spring which more or less maintains its form and movement, but in which the water **molecules** are always different.

C

Thus ageing and death should not be seen as **inevitable**, particularly as the **organism** **possesses** many **mechanisms** for repair. It is not, in principle, necessary for a biological system to age and die. Nevertheless, a restricted life **span**, ageing, and then death are basic characteristics of life. The reason for this is easy to recognise: in nature, the **existent** organisms either **adapt** or are regularly replaced by new types. Because of changes in the genetic material (**mutations**) these have new characteristics and in the course of their **individual** lives they are tested for **optimal** or better **adaptation** to the environmental conditions. **Immortality** would **disturb** this system - it needs room for new and better life. This is the basic problem of **evolution**.

D

Every **organism** has a life **span** which is highly characteristic. There are **striking** differences in life **span** between different **species**, but within one **species** the **parameter** is relatively constant. For example, the **average duration** of human life has hardly changed in thousands of years. Although more and more people **attain** an advanced age as a result of developments in **medical** care and better nutrition, the characteristic **upper** limit for most **remains** 80 years. A **further** argument against the simple **wear and tear theory** is the **observation** that the time within which organisms age lies between a few days (even a few hours for **unicellular** organisms) and several thousand years, as with **mammoth** trees.

E

If a life **span** is a genetically **determined** biological characteristic, it is logically necessary to **propose** the existence of an **internal clock**, which in some way measures and controls the ageing **process** and which finally determines death as the last step in a

destruction = damage, obliteration, demolition, #construction dɪˈstrʌkʃn
permanent = lasting, never-ending, everlasting, eternal, #temporary ˈpɜːmənən
equilibrium = steadiness, balance, stability, #imbalance ˌiːkwɪˈlɪbrɪəm
material = substance, matter, objects məˈtɪəriəl
molecule = the smallest unit into which any substance can be divided without losing its own chemical nature. ˈmɒlɪkjʊː

inevitable = unavoidable, predictable, foreseeable, #avoidable. ɪnˈevɪtəbl̩
possess = have, own, hold. pəˈzes
existent = in existence, extant, current ɪgˈzɪstənt
mutation = a change in the genetic structure of any animal or plant that makes it different from others of the same kind. mjuːˈteɪʃn
optimal = optimum, goal, ideal, best-case scenario ˈɒptɪml̩
adaptation = alteration, adjustment, modification, change ædæpˈteɪʃn
immortality = the state of living forever or being remembered forever. ɪmɔːˈtælɪti
disturb = interrupt, bother, distract. dɪˈstɜːb
evolution = development, progress, progression. ɪˈvəːluːʃn

striking = outstanding, prominent, noticeable ˈstraɪkɪŋ
parameter = limitation, boundary, restriction. pəˈræmɪtə
attain = reach, achieve, get. əˈteɪn
upper = higher, better, greater. ˈʌpə
wear and tear theory = aging theory. weər ənd teə ˈθiəri
observation = surveillance, scrutiny, #neglect ɒbzəˈveɪʃn
unicellular = consisting of only one cell juːnɪˈseljələ
mammoth = enormous, massive, immense, huge, #tiny ˈmæməθ

determine = decide, conclude, settle on dɪˈtɜːmɪn
propose = suggest, recommend, offer prəˈpəʊz
internal clock = biological clock ɪnˈtɜːnl̩ ˈklɒk

fixed programme. Like the life span, the **metabolic rate** has for different organisms a fixed mathematical relationship to the **body mass**. In comparison to the life span this relationship is 'inverted': the larger the organism the lower its metabolic rate. Again this relationship is **valid** not only for birds, but also, similarly on average within the **systematic** unit, for all other organisms (plants, animals, unicellular organisms).

F

Animals which behave '**frugally**' with energy become particularly old, for example, crocodiles and tortoises. Parrots and birds of prey are often held **chained up**. Thus they are not able to 'experience life' and so they attain a high life span in **captivity**. Animals which save energy by **hibernation** or **lethargy** (e.g. bats or hedgehogs) live much longer than those which are always active. The metabolic rate of mice can be reduced by a very low consumption of food (hunger diet). They then may live twice as long as their well fed **comrades**. Women become **distinctly** (about 10 per cent) older than men. If you examine the metabolic rates of the two sexes you establish that the higher male metabolic rate **roughly** accounts for the lower male life span. That means that they live life '**energetically**' - more **intensively**, but not for as long.

G

It follows from the above that **sparing** use of energy reserves should tend to extend life. Extreme high performance sports may lead to optimal **cardiovascular** performance, but they quite certainly do not **prolong** life. Relaxation lowers metabolic rate, as does **adequate** sleep and in general an **equable** and balanced personality. Each of us can develop his or her own 'energy saving programme' with a little self-observation, **critical** self-control and, above all, logical consistency. Experience will show that to live in this way not only increases the life span but is also very healthy. This final aspect should not be forgotten.

metabolism = the chemical processes by which food is changed into energy in your body.

mə'tæbə,lizəm

body mass index = BMI = is a value derived from the mass (weight) and height of a person.

'bɒdi,mæs'ɪndeks

invert = turn upside down, turn over, double back. ɪn'vɜ:t

valid = is legally or officially acceptable >> invalid 'vælɪd

systematic = methodical, organized, #disorganized sɪstə'mætɪk

frugal = careful, cautious, sparing, #extravagant 'fru:ɡl

chain up = capture, bind, manacle = to fasten someone or something to something else using a chain, especially in order to prevent them from escaping or being stolen 'tʃeɪn ʌp

captivity = imprisonment, confinement, #freedom kæp'tɪvɪti

hibernate = if an animal hibernates, it sleeps for the whole winter. 'haɪbəneɪt

lethargy = weariness, tiredness, #energy. 'leθədʒi

comrade = companion, friend, buddymate. 'kɒmreɪd

distinctly = clearly, noticeably, definitely. dɪ'stɪŋktli

roughly = approximately, about, around, #exactly 'rʌfli

energetically = actively, dynamically, powerfully. enə'dʒetɪkli

intensive = concentrated, exhaustive, thorough. ɪn'tensɪv

sparing = using very little of something 'speərɪŋ

reserve = keep, save, preserve. rɪ'zɜ:v

cardiovascular = relating to the heart and blood vessels. kɑ:'diəʊ'væskjələ

prolong = lengthen, extend, make longer, drag something out #curtail prə'lɒŋ

adequate = sufficient, enough, #inadequate. 'ædɪkwət

equable = someone who is equable remains calm and happy and does not often get annoyed. 'ekwəbəl

critical = significant, vital, important. 'krɪtɪkəl

TEST 4

READING PASSAGE 1

Land Of The Rising Sun



A

Japan has a **significantly** better record in terms of **average** mathematical **attainment** than England and Wales. Large **sample** international comparisons of pupils' attainments since the 1960s have established that not only did Japanese pupils at age 13 have better scores of **average attainment**, but there was also a larger **proportion** of 'low' attainers in England, where, **incidentally**, the **variation** in attainment scores was much greater. The percentage of Gross National Product spent on education is reasonably **similar** in the two countries, so how is this higher and more **consistent** attainment in maths achieved?

B

Lower secondary schools in Japan cover three school years, from the seventh **grade** (age 13) to the ninth **grade** (age 15). **Virtually** all pupils at this stage **attend** state schools: only 3 per cent are in the private **sector**. Schools are usually modern in design,

significantly = considerably, significantly,
#insignificantly. sig'nɪfɪkəntli

average = calculated by adding several amounts together, finding a total, and dividing the total by the number of amounts 'ævərɪdʒ

attainment = achievement, accomplishment, fulfillment, #failure ə'teɪnmənt

incidentally = in a way that was not planned but that is connected with something else ɪn'sɪdəntəli

consistent: constant, stable, steady, #inconsistent kən'sɪstənt

virtually = almost, nearly, not quite, practically 'vɜ:tʃʊəli

attend = appear, take part in, enroll, go to ə'tend

sector = division, area, zone 'sektə

set well back from the road and **spacious** inside. Classrooms are large and pupils sit at single desks in rows. Lessons last for a standardised 50 minutes and are always followed by a 10-minute break, which gives the pupils a chance to **let off steam**. Teachers begin with a formal **address** and **mutual bowing**, and then **concentrate** on whole-class teaching. Classes are large - usually about 40 - and are **unstreamed**. Pupils stay in the same class for all lessons throughout the school and develop **considerable** class **identity** and loyalty. Pupils **attend** the school in their **own** neighbourhood, which in **theory** removes ranking by school. In practice in Tokyo, because of the relative **concentration** of schools, there is some **competition** to get into the 'better' school in a **particular** area.

C

Traditional ways of teaching form the basis of the lesson and the **remarkably** quiet classes take their **own** notes of the points made and the examples **demonstrated**. Everyone has their **own** copy of the textbook supplied by the central education authority, Monbusho, as part of the **concept** of free **compulsory** education up to the age of 15. These textbooks are, **on the whole**, small, **presumably** inexpensive to produce, but well set out and logically developed. (One teacher was particularly **keen** to introduce colour and pictures into maths textbooks: he felt this would make them more **accessible** to pupils brought up in a cartoon culture.) Besides approving textbooks, Monbusho also decides the highly **centralised** national **curriculum** and how it is to be delivered.

D

Lessons all follow the same **pattern**. At the beginning, the pupils put solutions to the homework on the **board**, then the teachers comment, correct or **elaborate** as necessary. Pupils mark their **own** homework: this is an important **principle** in Japanese schooling as it **enables** pupils to see where and why they made a mistake, so that these can be avoided in future. No one minds mistakes or **ignorance** as long as you are prepared to learn from them. After the homework has been discussed, the teacher

spacious = airy, commodious, capacious, #cramped, narrow. 'speɪʃəs
let off steam = relax, unwind, let hair down 'let of sti:m
mutual = related = feeling the same emotion, or doing the same thing to or for each other 'mju:tʃʊəl
bowing = the act of bending the top part of your body forward to show respect for someone when you meet them 'bəʊɪŋ
unstreamed = to not be put into groups according to students' ability ʌn'stri:mɪd
considerable = significant, great, huge kən'sɪdərəbəl
identity = uniqueness, distinctiveness, characteristics aɪ'dentɪti
concentration = attention, focus, #distraction kɒnsən'treɪʃn
competition = contest, championship, tournament, quiz kəmpe'tɪʃn
particular = certain, precise, specific pə'tɪkjʊlə

remarkably = surprisingly, extraordinarily, outstandingly, #unremarkably ɪ'mɑ:kəbli
demonstrate: display, show, explain (monstra= show .i.e demonstrator) 'demənstreɪt
compulsory = obligatory, mandatory, required, #optional kəm'pʌlsəri
on the whole = generally, in general, all in all 'ɒn ðə həʊl
presumably = probably, seemingly, likely prɪ'zju:məbli
accessible = approachable, available, handy, reachable, #inaccessible (ac= toward or movement .i.e accelerate, action) ək'sesəbəl
centralise = to organize the control of a country, organization, or system so that everything is done or decided in one place (cen= middle .i.e center, centre) 'sentrəlaɪz
curriculum = subjects, program, course. kə'rɪkjʊləm

pattern = form, model, plan 'pætən
elaborate = say more, explain, give details, go into detail ɪ'læbəreɪt
principle = standard, idea, moral rule, belief 'prɪnsəpəl
enable = aid, assist, support, facilitate, # prevent ɪ'neɪbəl
ignorance = unawareness, inexperience, unintelligence, lack of knowledge or information about something. 'ɪgnərəns

explains the topic of the lesson, slowly and with a lot of **repetition** and **elaboration**. Examples are demonstrated on the board; questions from the textbook are **worked through** first with the class, and then the class is set questions from the textbook to do individually. Only rarely are **supplementary** worksheets **distributed** in a maths class. The impression is that the logical nature of the textbooks and their **comprehensive coverage** of different types of examples, combined with the relative **homogeneity** of the class, **renders** work sheets unnecessary. At this point, the teacher would **circulate** and make sure that all the pupils were coping well.

E

It is remarkable that large, mixed-ability classes could be kept together for maths throughout all their **compulsory** schooling from 6 to 15. Teachers say that they give **individual** help at the end of a lesson or after school, **setting** extra work if necessary. In **observed** lessons, any **strugglers** would be **assisted** by the teacher or quietly **seek** help from their neighbour. Carefully **fostered** class identity makes pupils **keen** to help each other - anyway, it is in their interests since the class **progresses** together. This **scarcely** seems **adequate** help to **enable** slow learners to **keep up**. However, the Japanese **attitude** towards education runs along the lines of 'if you work hard enough, you can do almost anything'. Parents are kept closely informed of their children's progress and will **play a part in** helping their children to keep up with class, sending them to 'Juku' (private evening **tuition**) if extra help is needed and encouraging them to work harder. It seems to work, at least for 95 per cent of **the school population**.

repetition = reiteration, repeating, replication
repi'tiʃn

elaboration = illustration, amplification, explanation
ɪ,læbə'reiʃn

work through = to manage a problem that has many different parts step by step
'wɜ:k θru:

supplementary = additional, extra, added (*ple=fill, full .i.e plenty, replete*)
sʌplɪ'mentri

distribute = allocate, dispense, spread
dɪ'strɪbjʊt
comprehensive = complete, far-reaching, wide-ranging, #incomplete, #sketchy (*com= together .i.e combine, complete*)
kɒmprɪ'hensɪv

coverage = attention, reportage, reporting
'kʌvərɪdʒ

homogeneity = consistency, regularity, #unevenness (*hom=same .i.e homogeneous*)
həʊmədʒɪ'nɪəti

render = make, leave, cause to be/become
'rendə

circulate = mingle, move around, communicate. (*circ= circle .i.e circus, circular*)
'sɜ:kjʊleɪt

individual = personal, private, specific, for one person
ɪndɪ'vɪdʒʊəl

observe = study, see, notice, witness
əb'zɜ:v

struggle = fight, effort, strive

strugglers = those who struggle
'strʌgl

assist = help, aid, support, help out, give somebody a hand, lend a hand
ə'sɪst

seek = search for, look for, find
sɪ:k

foster = encourage, promote, cultivate, #discourage
'fɒstə

progress = development, improvement, growth
prə'gres

scarcely = barely, hardly, just
'skeəsli

adequate = enough, sufficient,

#inadequate, #insufficient
'ædɪkwət

enable = allow, permit, assist, facilitate, #prevent
ɪ'neɪbl

keep up = follow, catch up, continue
ki:p ʌp

play a part in = play a role in, involve in, take part in, participate in
pleɪ ə pɑ:t ɪn

tuition = education, teaching, schooling, instruction
tju:'ʃn

the school population = learners, students, pupils
ðə sku:l ,pɒpjʊ'leɪʃn

F

So what are the major contributing factors in the success of maths teaching? Clearly, attitudes are important. Education is valued greatly in Japanese culture; maths is recognised as an important compulsory subject throughout schooling; and the emphasis is on hard work **coupled with** a focus on accuracy.

Other relevant points relate to the supportive attitude of a class towards slower pupils, the lack of competition within a class, and the positive emphasis on learning for oneself and improving one's own standard. And the view of repetitively boring lessons and learning the facts by heart, which is sometimes **quoted** in **relation** to Japanese classes, may be **unfair** and **unjustified**. No poor maths lessons were observed. They were mainly good and one or two were **inspirational**.

emphasis = focus, stress, prominence, highlighting 'emfəsis

couple with = combine, link with/to 'kʌp| wið

accuracy = correctness, precision, exactness, #inaccuracy 'ækjərəsi

relevant = related, appropriate, #unrelated rɪ'leɪvənt

quote = recite, repeat, refer to kwəʊt

relation = connection, association, link rɪ'leɪʃən

unfair = unjust, unequal, inequitable, biased ʌn'feə

unjustified = unfair, unwarranted, #justified ʌn'dʒʌstɪfaɪd

inspirational = providing encouragement or new ideas for what you should do = motivational ɪnspə'reɪʃənl

READING PASSAGE 2

Biological control of pests

The continuous and **reckless** use of **synthetic**

chemicals for the control of pests which **pose** a threat to agricultural crops and human health is proving to be **counter-productive**. Apart from **engendering** **widespread** ecological **disorders**, **pesticides** have contributed to the emergence of a new **breed** of chemical-**resistant**, highly **lethal** **superbugs**.

According to a recent study by the Food and Agriculture Organisation (FAO), more than 300 **species** of agricultural pests have developed **resistance** to a wide **range** of **potent** chemicals. Not to be left behind are the disease-spreading pests, about 100 **species** of which have become **immune** to a variety of **insecticides** now in use.

One glaring disadvantage of pesticides' application is that, while destroying harmful pests, they also **wipe**

pest = bug, insect, vermin = a small animal or insect that destroys crop or food supplies. **pest**
reckless = irresponsible, thoughtless, careless, #cautious 'rekles

synthetic = artificial, manmade, manufactured sin'θetik

counter-productive = achieving the opposite result to the one that you want. 'kauntə-prə'daktiv

engender = produce, cause, create, stimulate, provoke in'dʒendə

disorder = illness, disease, infection dis'ɔ:də

lethal = deadly, dangerous, harmful,

#life-giving 'li:θl

superbug = a type of bacteria that cannot be killed by traditional drugs. 'su:pəbʌg

resistance = fight, battle, confrontation,

#surrender rɪ'zɪstəns

potent = powerful, strong, effective, influential (pot= power .i.e **potential**, **despot**) 'pəʊnt

immune = resistant, insusceptible, invulnerable, #susceptible i'mju:n

insecticide = pesticide, insect repellent, bug juice, fly spray (cid=kill) in'sektɪsaɪd

wipe out = destroy, eradicate, obliterate, remove, devastate, #protect waɪp 'aʊt

out many useful non-targeted **organisms**, which keep the growth of the pest population in check. This results in what **agroecologists** call the 'treadmill syndrome'. Because of their **tremendous** breeding potential and genetic diversity, many pests are known to **withstand** synthetic chemicals and **bear offspring** with a **built-in** resistance to pesticides.

The **havoc** that the 'treadmill syndrome' can bring about is well illustrated by what happened to cotton farmers in Central America. In the early 1940s, **basking** in the **glory** of chemical-based intensive agriculture, the farmers **avidly** took to pesticides as a sure measure to boost crop yield. The insecticide was applied eight times a year in the mid-1940s, rising to 28 in a season in the mid-1950s, following the sudden **proliferation** of three new varieties of chemical-resistant pests.

By the mid-1960s, the situation took an **alarming** turn with the **outbreak** of four more new pests, **necessitating** pesticide spraying to such an extent that 50% of the financial **outlay** on cotton production was accounted for by pesticides. In the early 1970s, the spraying frequently reached 70 times a season as the farmers were pushed to the wall by the **invasion** of **genetically** stronger insect species.

Most of the pesticides in the market today remain **inadequately** tested for **properties** that cause cancer and **mutations** as well as for other **adverse** effects on health, says a study by United States environmental agencies. The United States National Resource Defense Council has found that DDT was the most popular of a long list of dangerous chemicals in use.

In the face of the **escalating** **perils** from **indiscriminate** applications of pesticides, a more effective and ecologically sound strategy of biological control, involving the selective use of natural enemies

organism = an animal, plant, human or any other living thing. 'ɔ:gənɪzəm
agroecologist = a person who is specialized in the study of ecological processes applied to agricultural production systems 'agrəʊɪ'kɒlədʒɪst
tremendous = huge, massive, enormous trɪ'mendəs
withstand = resist, endure, survive, tolerate wɪð'stænd
bear = produce, give birth to, bring into being bi:
offspring = descendants, children, progeny 'ɒfsprɪŋ
built-in = natural, innate, intrinsic bɪlt-'ɪn

havoc = chaos, mayhem, #order 'hævək
treadmill = routine, drudgery, grindstone 'tredmɪl
syndrome = condition, disease, set of symptoms 'sɪndrəʊm
bask = enjoy, savor, relish, luxuriate bɑ:sk
glory = admiration, prestige, honour 'glɔ:ri
avidly = keenly, enthusiastically, eagerly, #indifferently 'ævɪdli
yield = produce, generate, harvest ji:ld
proliferation = explosion, abundance, overprovision prə'lfə'reɪʃn

alarming = frightening, shocking, #calming ə'la:mɪŋ
outbreak = eruption, epidemic, outburst 'aʊtbreɪk
necessitate = require, demand, need nɪ'sesɪteɪt
outlay = spending, expenditure, costs, expenses, outgoings, outlay, overheads. 'aʊtler
invasion = attack, raid, arrival, #withdrawal ɪn'veɪʒn
genetically = innately, natively, naturally dʒɪ'netɪklɪ

inadequately = poorly, insufficiently, improperly ɪn'ædɪkwətli
property = material goods, belongings, stuff 'prɒpəti
mutation = a change in the genetic structure of an animal or plant that makes it different from others of the same kind mju:'teɪʃn
adverse = not good or favourable = negative and unpleasant 'ædvɜ:s

escalate = rise, soar, rocket, #plummet 'eskəleɪt
peril = danger, threat, risk, hazard, #safety 'perəl
indiscriminate = unselective, random, #selective ɪndɪ'skrɪmɪnət

of the pest population, is fast gaining popularity - though, as yet, it is a new field with limited potential. The advantage of biological control in contrast to other methods is that it provides a relatively low-cost, **perpetual** control system with a minimum of **detrimental** side-effects. When handled by experts, bio-control is safe, non-polluting and **self-dispersing**.

The Commonwealth Institute of Biological Control (CIBC) in Bangalore, with its global network of research laboratories and field stations, is one of the most active, non-commercial research agencies engaged in pest control by setting natural **predators** against **parasites**. CIBC also serves as a clearing-house for the export and import of biological agents for pest control world-wide.

CIBC successfully used a seed-feeding **weevil**, native to Mexico, to control the **obnoxious** parthenium weed, known to **exert** **devious** influence on agriculture and human health in both India and Australia. Similarly the Hyderabad-based Regional Research Laboratory (RRL), supported by CIBC, is now trying out an Argentinian weevil for the **eradication** of water **hyacinth**, another dangerous weed, which has become a **nuisance** in many parts of the world. According to Mrs Kaiser Jamil of RRL, 'The Argentinian weevil does not attack any other plant and a pair of adult bugs could destroy the weed in 4-5 days.' CIBC is also perfecting the technique for breeding parasites that prey on 'disapene scale' insects - **notorious** **defoliants** of fruit trees in the US and India.

How effectively biological control can be pressed into service is proved by the following examples. In the late 1960s, when Sri Lanka's **flourishing** coconut groves were **plagued** by leaf-mining **hispides**, a **larval** **parasite** imported from Singapore brought the pest under control. A natural predator **indigenous** to India, *Neodumetia sangawani*, was found useful in controlling the Rhodes grass-scale insect that was **devouring** **forage** grass in many parts of the US. By

perpetual = lasting, continual, frequently repeated, in a way that is annoying *pə'petʃuəl*
detrimental = harmful, damaging, negative, hazardous, pernicious *detri'mentl*
side-effect = unexpected result, consequence, knock-on effect *said- i'fekt*
disperse = scatter, disband, diffuse, break up *di'spɜ:s*

laboratory = workroom, test center, workshop *lə'bɒrətɹi*
predator = marauder, killer, hunter *'predətə*
parasite = a plant or animal that lives on or in another plant or animal and gets food from its. *'pærəsait*

weevil = a small insect that feeds on grain, flour etc and spoils it. *'wi:vɪl*
obnoxious = horrible, unpleasant, loathsome, #delightful *əb'nɒkʃəs*
exert = apply or bring to bear (a force/influence, or quality) *ig'zɜ:t*
devious = deceitful, underhanded, sly *'di:vɪəs*
eradicate = remove, get rid of, eliminate, erase *i'rædɪkeɪt*
hyacinth = a garden plant with blue, pink or white bell - shaped flowers and a sweet smell. *'haɪəsɪnθ*
nuisance = annoyance, bother, irritation *'nju:sns*
notorious = infamous, disreputable, tarnished, #famous *nəu'tɔ:riəs*
defoliant = a chemical substance, used especially in war, that makes all the leaves of plants drop off. *di:'fəʊliənt*

flourish = thrive, succeed, prosper, #deteriorate, #decline *'flaʊɹɪʃ*
plague = afflict, cause suffering to, trouble. *pleɪg*
hispide = large shrub or small tree of the eastern united states *'hɪspɪdʒ*
larval = adjective of "larva", which means young insect *'lɑ:vɪ*
parasite = a plant or animal that lives on or in another plant or animal and gets food from it *'pærəsait*
indigenous = native, original, aboriginal, local, #foreign *ɪn'dɪdʒɪnəs*
devour = demolish, consume, eat greedily, destroy (*de= removing .i.e decline, decrease*) *di'vaʊə*
forage = food, fodder, feed *'fɒrɪdʒ*

using *Neochetina bruci*, a **beetle** native to Brazil, scientists at Kerala Agricultural University **freed** a 12-kilometre-long **canal** from the **clutches** of the weed *Salvinia molesta*, popularly called 'African Payal' in Kerala. About 30,000 hectares of rice fields in Kerala are **infested** by this weed.

beetle = an insect with a round hard back that is usually black. ^{'bi:təl}
free – freed (past tense) = release = set free, discharge ^{fri:}
canal = waterway, seaway, inland waterway ^{kə'næl}
clutch = power, control. domination ^{klʌtʃ}
infest = invaded, filled, infected ^{ɪn'fest}

READING PASSAGE 3

Collecting ant specimens

Ancient voyagers who settled the far-flung collecting ants can be as simple as picking up stray ones and placing them in a jar, or as complicated as completing an exhaustive survey of all species present in an area and estimating their relative abundances. The exact method used will depend on the final purpose of the collections. For taxonomy, or classification, long series, from a single nest, which contain all castes (workers, including majors and minors, and, if present, queens and males) are desirable, to allow the determination of variation within species. For ecological studies, the most important factor is collecting identifiable samples of as many of the different species present as possible.

Unfortunately, these methods are not always compatible. The taxonomist sometimes

ancient = prehistoric, very old, earliest, #modern
'eɪnfənt

voyager = traveler, explorer, adventurer 'vɔɪdʒə

far-flung = far, distant, remote fɑː- 'flʌŋ

jar = pot, container, vessel dʒɑː

complicated = complex, intricate, convoluted
'kɒmplɪkeɪtɪd

exhaustive = thorough, comprehensive, in-depth ɪg'zɔːstɪv

abundance = plenty, wealth, profusion ə'bʌndəns

taxonomy, classification = taxonomic system, nomenclature, categorization tæk'sɒnəmi, klæsɪfɪ'keɪʃn

nest = a place made or chosen by a bird to lay its eggs in and to live in

caste = class, type, social order kɑːst

desirable = attractive, wanted, pleasing dɪ'zaɪərəbəl

identifiable = recognizable, distinguishable, classifiable aɪ'dentɪ'faɪəbəl

compatible = well-matched, well-suited, similar, #different, #incompatible (com=together .i.e combine)
kəm'pætəbəl

overlooks whole species in **favour of** those groups currently under study, while the ecologist often collects only a limited number of specimens of each species, **thus** reducing their value for taxonomic investigations.

To collect as wide a range of species as possible, several methods must be used. These include hand collecting, using **bait**s to attract the ants, ground **litter** sampling, and the use of **pitfall** traps. Hand collecting **consists of** searching for ants everywhere they are likely to **occur**. This includes on the ground, under rocks, logs or other objects on the ground, in **rotten** wood on the ground or on trees, in vegetation, on tree **trunks** and under **bark**. When possible, collections should be made from nests or **foraging columns** and at least 20 to 25 individuals collected. This will ensure that all individuals are of the same species, and so increase their value for detailed studies. Since some species are largely **nocturnal**, collecting should not be **confined** to daytime. Specimens are collected using an **aspirator** (often called a **pooter**), **forceps**, a fine, **moistened** paint brush, or fingers, if the ants are known not to **sting**. Individual insects are placed in plastic or glass tubes (1.5-3-0 ml capacity for small ants, 5-8 ml for larger ants) containing 75% to 95% ethanol. Plastic tubes with **secure tops** are better than glass because they are lighter, and do not break as easily if **mishandled**.

Baits can be used to attract and concentrate **foragers**. This often increases the number of individuals collected and attracts species that are **otherwise** elusive. Sugars and meats or oils will attract different species and a range should be **utilised**. These baits can be placed either on the ground or on the trunks of trees or large **shrubs**. When placed on the ground, baits should be **situated** on small paper cards or other flat, light-coloured surfaces, or in **test-tubes** or **vials**. This makes it easier to **spot** ants and to **capture** them before they can escape into the surrounding leaf litter.

overlook = ignore, miss, neglect, skip əʊvəˈlʊk
in favour of = if you are in favour of somebody/something, you support and agree with them/it ɪn ˈfeɪvər ɒv
thus = therefore, hence, as a result, accordingly ðʌs

bait = food used to attract fish, animals, or birds so that you can catch them. beɪt
litter = rubbish, trash, garbage ˈlɪtə
consist of = comprise, involve, be composed of kənˈsɪst ɒv
occur = happen, take place, strike əˈkɜː
rotten = (of food, wood, etc.) that has decayed and cannot be eaten or used ˈrɒtən
trunk = the thick central woody stem of a tree trʌŋk
bark = the outer covering of a tree bɑːk
foraging = hunting, searching, seeking ˈfɔːrɪdʒɪŋ
foraging column = a group of ants that finds food together ˈfɔːrɪdʒɪŋ ˈkɒləm
nocturnal = nighttime, nightly, #diurnal nɒkˈtɜːnl
confined = restricted, limited, narrowed kənˈfaɪnd
aspirator = an instrument or apparatus for aspirating fluid from a vessel or cavity ˈæspɪreɪtə
pooter = a bottle for collecting small insects and other invertebrates, having one tube through which they are sucked into the bottle and another, protected by muslin or gauze, which is sucked. ˈpuːtə
forceps = a medical instrument used for picking up and holding things. ˈfɔːseps
moisten = wet, dampen, moisturize, humidify ˈmɔɪsən
sting = bite, tingle, bite mark, puncture stɪŋ
mishandle = mismanage, misuse, mess up mɪsˈhændl

forager = the type of the ants that find food ˈfɔːrɪdʒə
otherwise = if not, or else, then ˈɒðəwaɪz
elusive = mysterious, intangible, vague, #obvious ɪˈluːsɪv
utilise = use, make use of something, employ ˈjuːtelaɪz
shrub = plant, herb, weed, bulb ʃrʌb
situated = placed, located, positioned ˈsɪtʃueɪtɪd
test-tube = a small glass container that is shaped like a tube and is used in chemistry test - tjuːb
vial = a very small bottle used for medicine, perfume etc ˈvaɪəl
spot = identify, notice, recognize spɒt
capture = catch, seize, trap ˈkæptʃə

Many ants are small and forage primarily in the layer of leaves and other **debris** on the ground. Collecting these species by hand can be difficult. One of the most successful ways to collect them is to **gather** the leaf litter in which they are foraging and extract the ants from it. This is most commonly done by placing leaf litter on a screen over a large **funnel**, often under some heat. As the leaf litter dries from above, ants (and other animals) move downward and eventually fall out the bottom and are collected in alcohol placed below the funnel. This method works especially well in rain forests and **marshy** areas. A method of improving the catch when using a funnel is to sift the leaf litter through a **coarse** screen before placing it above the funnel. This will concentrate the litter and remove larger leaves and **twigs**. It will also allow more litter to be sampled when using a limited number of funnels.

The **pitfall** trap is another commonly used tool for collecting ants. A pitfall trap can be any small container placed in the ground with the top level with the surrounding surface and filled with a **preservative**. Ants are collected when they fall into the trap while foraging.

The **diameter** of the traps can vary from about 18 mm to 10 cm and the number used can vary from a few to several hundred. The size of the traps used is influenced largely by personal **preference** (although larger sizes are generally better), while the number will be determined by the study being **undertaken**. The preservative used is usually ethylene glycol or propylene glycol, as alcohol will **evaporate** quickly and the traps will dry out.

One advantage of pitfall traps is that they can be used to collect over a period of time with minimal **maintenance** and **intervention**. One disadvantage is that some species are not collected as they either avoid the traps or do not commonly **encounter** them while foraging.

debris = remains, fragments, wreckage 'deɪbriː

gather = collect, group, get together, join together, #disperse 'gæðə

funnel = a thin tube with a wide top that you use for pouring liquid into a container with a narrow opening, such as a bottle. 'fʌn]

marshy = muddy, wet, boggy, #dry (mar=water, sea .i.e marine, submarine) 'mɑːʃi

sift = sieve, filter, separate sɪft

coarse = rough, uneven, bumpy, rugged kɔːs

twig = a small very thin stem of wood that grows from a branch on a tree. twɪg

pitfall = a problem or difficulty that is likely to happen in a particular job, course of action, or activity. 'pɪtfɔːl

preservative = protective, conserving, #destructive (serv=protect .i.e preserve, conserve) prɪ'zɜːvətɪv

diameter = width, length, breadth daɪ'æmɪtə

vary = differ, diverge, fluctuate 'veəri

preference = favorite, first choice, #indifference 'prefrəns

undertake = carry out, conduct, take on ʌndə'teɪk

evaporate = if a liquid evaporates, or if heat evaporates it, it changes into a gas. ɪ'væpəreɪt

maintenance = preservation, continuation, protection, #destruction 'meɪntənəns

intervention = interference, intrusion, involvement ɪntə'venʃn

encounter = meet, come across, stumble upon ɪn'kaʊntə